

Almost-Total Puzzles and Their Applications

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Abstract. Public-coin protocols are cryptographic protocols in which all messages sent by a specific party (typically the receiver or verifier) consist solely of random bits. These protocols have been extensively studied *in the classical setting* due to their advantageous properties in several scenarios, such as the parallel repetition of interactive arguments, and the design of secure multi-party computation with low round complexity, among others. Curiously, *post-quantum constructions* of public-coin protocols remain limited, particularly when optimization is sought in additional factors like round complexity or hardness assumptions.

We introduce the concept of *almost-total puzzles*, a novel cryptographic primitive characterized by two key properties: (i) hardness against any efficient adversary, and (ii) an “almost-total” guarantee of the existence of solutions, even when the puzzle generator is malicious. We demonstrate that this primitive can be derived from one-way functions in public-coin, requiring only two rounds. By leveraging this primitive, we obtain a family of new *public-coin* results in both the classical and post-quantum settings, based on the *minimal assumption* of (post-quantum) one-way functions, including:

- five-round post-quantum extractable commitments and witness-indistinguishable arguments of knowledge, where the (knowledge) extractors achieve the *coherently expected quantum-polynomial-time* (EQPT_c) simulation proposed by Lombardi, Ma, and Spooner [FOCS’22];
- five-round classical extractable commitments that *do not suffer from over extraction*;
- five-round classical delayed-input strong witness-indistinguishable arguments of knowledge, and delayed-input witness-hiding arguments of knowledge;
- the five-round post-quantum analogue of the last item, but with the difference that (1) the input can be delayed until the third round, and (2) post-quantum arguments of knowledge are again defined w.r.t. EQPT_c-simulation;
- $O(\log^* \lambda)$ -round post-quantum non-malleable commitments.

Keywords: Post-Quantum · Public-Coin · Extraction · Simulation · Round Complexity

* The work described in this paper was partially supported by a grant from the Research Grants Council of the Hong Kong Special Administrative Region, China [Project No.: CUHK 24208025].

† Supported in part by AFOSR Award FA9550-24-1-0156, NSF CAREER Award 2144303, NSF grants 2106263, and 2028920. Any opinions, findings and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the United States Government, AFOSR, or NSF.

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1 Introduction

Public-coin protocols are cryptographic schemes in which all messages from a specific party, typically the verifier, consist solely of random bits. Such protocols offer distinct advantages over private-coin protocols in certain scenarios. For instance, parallel repetition can reduce the soundness error, at an exponential speed, for public-coin *interactive arguments* of an arbitrary (polynomial) number of rounds [HPWP10, CL10, CP15]. In contrast, [BIN97] constructed a four-round *private-coin* interactive argument for which parallel repetition does not reduce the soundness error at all. Another key example is the renowned Fiat-Shamir transform [FS87], which applies exclusively to public-coin protocols. Additional advantages of the public-coin property are found in the literature on non-malleability and (concurrent) secure multi-party computation, particularly in the low-round-complexity settings (e.g., [GGJS12, COSV16, COSV17a, COSV17b, GKP17, CCG⁺20]), where the public-coin property enables rewinding-based techniques in settings where rewinding opportunities are very limited due to round constraints.

Given these valuable properties, public-coin protocols have been extensively studied *in the classical settings*. Public-coin constructions, even round-optimal and from minimal assumptions, are known for foundational cryptographic primitives such as extractable commitments (ExtCom) [PW09], non-malleable commitments (NMCom) [GPR16, GR19] and witness-indistinguishable (WI) proofs [Blu86, FS90a]. Furthermore, a seminal body of work explores the limitations of zero-knowledge protocols, articulating an “impossible trinity” among constant-round complexity, black-box simulators, and public-coin verifiers [GK90, Bar01, BL02].

The Curious Post-Quantum Case. In contrast, public-coin protocols remain underexplored⁵ in the so-called *post-quantum* setting, where honest parties and their communication channels are entirely classical, but the adversary is granted quantum computing power. Post-quantum protocols offer significant efficiency benefits, as they do not require honest parties to perform quantum operations. Also, they ensure that even if adversaries gain early access to quantum computing capabilities, the honest parties need not adopt quantum technologies to maintain security.

In recent years, significant progress has been made in post-quantum simulation and extraction techniques. However, somewhat unexpectedly, most of the proposed solutions rely on private-coin property, while public-coin constructions and techniques remain relatively underdeveloped, as we will elaborate next.

On Zero-Knowledge. Watrous [Wat06] constructed the first public-coin post-quantum zero-knowledge (PQ-ZK) protocol in a polynomial⁶ number of rounds. On the other hand, [CCLY22b] demonstrated the impossibility of constant-round public-coin PQ-ZK,⁷ unless one resorts to *non-black-box simulation* or assumes that $\mathbf{NP} \subseteq \mathbf{BQP}$. However, curiously, existing non-black-box PQ-ZK protocols [BS20, ABG⁺21] remain private-coin. This stands in contrast to the classical setting, where Barak [Bar01] presented a method for achieving public-coin ZK in constant rounds via non-black-box simulation, complementing the impossibility result of [GK90].

⁵ We remark that the claim here applies only to the plain model, where no trusted setup is available. There are studies that focus on lifting classical results to the post-quantum regime using trusted setups, such as (quantum-accessible) random oracles, which represents a different direction from the focus of the current work.

⁶ In fact, it can be optimized to $\omega(1)$ number of rounds. We can do $O(\log \lambda)$ times parallel repetition to reduce soundness error to $1/\text{poly}(\lambda)$ and then do $\omega(1)$ times sequential repetition to reduce the soundness error to $\text{negl}(\lambda)$.

⁷ Indeed, this (public-coin) impossibility holds even for PQ-ZK *with ε -simulation* which is a relaxed notion of ZK that allows for an arbitrarily small noticeable simulation error ε .

On Simulatable Extraction. For primitives defined with *explicit* (knowledge) extractors, we emphasize that there are two different definitions in the (post-)quantum setting. A naive definition is to simply require the extractor to extract the secret from the malicious party successfully (e.g., [Unr12, CMSZ22]). Alternatively, a stronger definition will, in addition to the “successful extraction” guarantee, require the extractor to simulate the post-extraction view of the malicious party (e.g., [CCLY22a]).

In the classical setting, these two definitions are essentially equivalent. However, in the (post-)quantum setting, achieving the stronger notion, referred to as *simulatable extraction*, is surprisingly non-trivial due to the quantum no-cloning theorem — it requires substantial technical ideas to enable extraction without disturbing the adversary’s internal *quantum* state.

We emphasize that the stronger notion of simulatable extraction is more desirable and serves as the correct quantum analogue of the classical extraction requirement. This is particularly important because primitives such as extractable commitments and arguments of knowledge are often used as building blocks in larger protocols, such as secure two-party computation (2PC), where simulating the adversary’s post-extraction view is often crucial for ensuring overall security (e.g., in constructing the 2PC simulator).

The current state-of-the-art results w.r.t. simulatable extraction are summarized in Table 1 (see Rmk. 1 for notations). Notably, the construction of public-coin witness-indistinguishable arguments of knowledge (WIAoK) or ExtCom with standard simulation remains an open problem, *even without restrictions on (polynomial) round complexity*. If we relax the simulation requirement to EQPT_c-simulation (see Sec. 3.2 for a more detailed comparison), constant-round constructions are known. However, these results are either private-coin or rely on assumptions stronger than the minimal, i.e., post-quantum one-way functions (PQ-OWFs). This intriguing state-of-the-art scenario raises fundamental questions:

Question: *In the post-quantum regime, is it possible to have public-coin simulatable extraction from the minimal assumption of PQ-OWFs? Is it possible if we relax the simulation requirement to EQPT_c-simulation or ε -simulation?*

Table 1: Summary of Post-Quantum Simulatable Extraction (See Rmk. 1)

	Citation	Round Complexity	Public-Coin	Simulation	Assumption
WIAoK	[ACL21, Thm. 2]	$\text{poly}(\lambda)$	✗	standard	QLWE
ExtCom	[ABG ⁺ 21, Sec. 5]	7	✗	standard	QFHE and more
WIAoK & ExtCom	[LMS22, Thm. 1.9, Cor. 1.10]	4	✓	EQPT _c	Collapsing Hash
	[LMS22, Thm. 1.9, Cor. 1.10]	4	✓	EQPT _c	super-poly OWF
	[LMS22, Thm. 1.9, Cor. 1.10]	3	✓	EQPT _c	super-poly NIC
	[LMS22, Cor. 2.7]	12	✗	EQPT _c	OWF
	Corollary 1, this work	5	✓	EQPT _c	OWF

Remark 1. Some remarks regarding Table 1 follow:

1. It should be clear that all results presented are in the post-quantum setting, though for brevity, we omit the prefix “PQ-” in front of terms like WIAoK, OWF, etc.

2. EQPT_c -simulation, introduced in [LMS22], is a relaxed simulation notion. It allows the simulator to *coherently* execute expected QPT procedures, resulting in a superposition of computations with different runtimes, and then “revert” the runtime by executing the computation in reverse. It is worth mentioning that EQPT_c simulation implies ε -simulation, which is a widely accepted notion with interesting implications, such as WI and IIC (Input-Indistinguishable Computation). Therefore, all of our results remain valid and interesting even if we replace the term “ EQPT_c -simulatable” with “ ε -simulatable” in all relevant theorems herein.
3. “NIC” in the last column of the fifth row stands for “non-interactive commitment.”
4. It only includes results that were *explicitly* stated in the literature. For example, [ABG⁺21] built a PQ-MPC and thus must implicitly build a WIAoK as well; but we choose not to include such indirect results. This choice does not change the main message Table 1 is meant to convey.
5. Some rows may only contain a representative result of the same type. For example, [CCLY22a] contains similar results as the sixth row in Table 1; [HSS11, LN11] contain similar results as the first row in Table 1. We choose not to include them.

1.1 Our Results

In this work, we introduce a new primitive that we call *almost-total puzzles* (ATPs). To understand this notion, consider an interaction between a *sender* G and a *receiver* R . At the end of their interaction, G and R agree on a puzzle. This puzzle is required to satisfy the following two properties:

- **Hard:** the generated puzzle must be hard to solve for any *efficient* R , *even if* R *behaves maliciously while interacting with* G .
- **Almost-total:** a solution for the generated puzzle is guaranteed to exist except for with negligible probability, *even if* G *behaves maliciously while interacting with* R .

Intuitively, the notion of ATP is useful because it sets up a hard problem for which a “trapdoor” (i.e., solution) is guaranteed to exist. As a veteran may already notice, this represents a versatile gadget that would be useful in settings like the Feige-Lapidot-Shamir (FLS) paradigm [FS90a, FS90b, FLS90]. It is worth mentioning that we do not require (even the honest) sender G to know the solution to the generated puzzle. This is because, for typical applications such as [FS90a, FS90b, FLS90, Pas03], the mere existence of the solution already suffices; and this indeed applies to all the scenarios considered in this work.

As will be demonstrated shortly, ATPs are particularly useful when they are *public-coin* and *round-efficient*. Fortunately, we obtain such ATPs from the minimal assumption of OWFs, in both the classical and post-quantum settings:

Theorem 1 (Informal). *Assuming the existence of OWFs (resp. PQ-OWFs), there exists a two-round, public-coin construction of ATPs (resp. post-quantum ATPs).*

It is also worth mentioning that when we restrict an ATP to a single round and require the existence of a solution with probability 1 (as opposed to merely overwhelming probability), then it is indeed equivalent to the existence of hard-on-average **TFNP** problems.⁸ It is well-established that the average-case hardness of **TFNP** can be derived from the existence of one-way permutations or collision-resistant hash functions. Furthermore, [HNY17] demonstrated that the existence

⁸ **TFNP** stands for the complexity class of “total-function” **NP**, introduced by Megiddo and Papadimitriou [MP91]. It is a subclass of **FNP** with the additional property of the problem being *total*, i.e., for any instance a solution is guaranteed to exist.

of OWFs implies the average-case hardness of **TFNP**, under an additional derandomization assumption. Whether this derandomization assumption can be eliminated remains an open question (see [HKKS20, FGH⁺23]).

In this context, [Thm. 1](#) can be interpreted as substituting the derandomization assumption with interactions. Specifically, [Thm. 1](#) represents a relaxed version of hard-on-average **TFNP** that permits public-coin interaction while allowing for a negligible probability of no solution existing. As we will now demonstrate, this relaxed version leads to several novel results, both in the classical and post-quantum regimes.

1.1.1 Application I: Post-Quantum Simulatable Extraction in Public-Coin

As the first application, the post-quantum almost-total puzzles (PQ-ATPs) from [Thm. 1](#) contribute positively towards addressing the earlier [Question](#) regarding public-coin simulatable extraction. In particular, we demonstrate that, at the cost of three extra rounds, public-coin PQ-ATPs imply public-coin PQ-ExtCom and PQ-WIAoK, both with EQPT_c-simulation. In light of [Thm. 1](#), we can derive the following corollary:

Corollary 1. *Assuming the existence of PQ-OWFs, there exist a PQ-ExtCom and a PQ-WIAoK, both being five-round, public-coin, and EQPT_c-simulatable.*

We remark that the (relaxed) EQPT_c simulation notion in [Corollary 1](#) is necessary for achieving *constant-round* PQ-WIAoK and PQ-ExtCom. This necessity arises from the aforementioned impossibility result regarding constant-round PQ-ZK protocols [CCLY22b], as well as the fact that both PQ-WIAoK (see [LMS22, Section 12]) and PQ-ExtCom (see [CCLY22a, Appendix A]) imply PQ-ZK.

A noteworthy aspect of [Corollary 1](#) is that, as shown in [Table 1](#), prior to this work, it remained unclear whether one could construct OWF-based PQ-ExtCom/PQ-WIAoK in public-coin, *even when allowing a polynomial number of rounds*.

1.1.2 Application II: (Classical) Public-Coin ExtCom without Over Extraction

The construction of PQ-ExtCom from PQ-OWF in [Corollary 1](#) also applies in the classical setting. Specifically, the same construction yields a classical ExtCom base on OWFs. Notably, the resulting ExtCom does not suffer from *over-extraction*. Over-extraction means that an extractor may extract some non- $\perp$ message from an invalid commitment instead of detecting the invalidness of the commitment (i.e., when the commitment has no valid opening).

In the classical setting, the issue of over-extraction has been observed in several prior works [GLOV12, GGJS12, Kiy14]. In fact, achieving ExtCom *without over extraction* is not particularly challenging if one is open to private-coin constructions. However, insisting on a public-coin construction necessitates stronger assumptions or super-constant rounds. Remarkably, this problem remained unresolved prior to the current work. That is, our construction is the first OWF-based ExtCom that *simultaneously* achieves public-coin and constant-round properties without over-extraction. We find this result interesting on its own:

Corollary 2. *Assuming the existence of OWFs, there exists a five-round, public-coin construction of extractable commitment that does not suffer from over-extraction.*

1.1.3 Application III: Strong WI and Friends

Strong witness indistinguishability (sWI) is an augmented notion of WI, which requires that for two indistinguishable statement distributions $\mathcal{X}_1, \mathcal{X}_2$, a proof (or argument) for statement $x_1 \leftarrow \mathcal{X}_1$

must be indistinguishable from that for statement $x_2 \leftarrow \mathcal{X}_2$. In this work, we focus on strong WI in the *delayed-input* setting [JKKR17], where the instance is determined by the prover only in the last round of the interaction. This delayed-input notion is incomparable but yet as useful as the original notion of strong WI. For scenarios where it requires to enforce “honest behavior” on participants and only requires indistinguishability-based security, (delayed-input) strong WI can be used to replace ZK. Such a replacement could benefit from a lower round complexity, bypassing impossibility results on ZK like [GK90, BL02].

For delayed-input strong WI, [JKKR17] proposed two constructions—a two-round construction based on quasi-polynomial-time secure oblivious transfers (OTs), and a three-round construction from polynomial-time secure OTs. The requirement for super-polynomial hardness in the two-round construction appears to be necessary, as noted by [Kiy21]. Furthermore, although not explicitly discussed in the literature, we observe that it is not hard to get a four-round construction from the average-case hardness of **TFNP**. Interestingly, in this low-round-complexity regime, no constructions are known from the minimal assumption of OWFs, *particularly with the public-coin requirement*. To address this gap, we present a five-round construction of delayed-input strong WI, from the existence of OWFs. Notably, our construction is public-coin, in contrast to both constructions in [JKKR17], which are private-coin.

Our result is actually stronger—we introduce a new variant of zero-knowledge called *delayed-input non-uniform ZK* (dinuZK). This notion differs from standard ZK in two key aspects: (1) the ZK property holds against a delayed-input (malicious) verifier, and (2) the simulator is a non-uniform machine that can depend on the verifier. This notion of dinuZK serves as a versatile alternative to standard ZK for “enforcing honest behavior” in scenarios where indistinguishability-based (IND-based) security suffices (instead of requiring the “full” simulation power of ZK). For instance, it is not hard to see that dinuZK implies both delayed-input strong WI and delayed-input WH.

We construct a compiler that converts ATPs (resp. public-coin ATPs) to dinuZK (resp. public-coin dinuZK), at the cost of three extra rounds. This compiler, combined with [Thm. 1](#), leads to the following result:

Corollary 3. *Assuming the existence of OWFs, there exists a five-round, public-coin construction of dinuZKAoK (and thus a delayed-input strong WIAoK and a delayed-input WHAoK).*

Post-Quantum Setting. Ideally, one would hope for an analogous result to [Corollary 3](#) in the post-quantum setting. But this appears challenging (see [Sec. 2.2](#) for related discussion). Nevertheless, we introduce yet another ZK variant called *post-quantum preprocessing ZKAoK* (PQ-ppZKAoK). This variant operates in two stages: a *preprocessing stage* and a *proof stage*, where the statement only becomes available at the beginning of the proof stage. It is worth mentioning that this preprocessing variant differs from the original delayed-input notion solely in how late the instance can be delayed—if the instance can be delayed until the last round, then it is the original delayed-input protocols; if the instance can only be delayed until some rounds before the last, it corresponds to our notion of preprocessing protocols. Regarding the notion of post-quantum AoK, we require knowledge extraction with EQPT_c simulation (see [Def. 9](#) for a formal definition).

Similar to the aforementioned dinuZK, PQ-ppZKAoK is valuable because it implies post-quantum WH and strong WI, *where the instance can be delayed until the beginning of the “on-line” (i.e., proof) stage*. We remark that these preprocessing variants could be just as useful as the (original) delayed-input protocols. This is due to the canonical “non-uniform” way in which delayed-input protocols are often utilized, as further discussed in the technical overview.

We present a compiler that converts (public-coin) PQ-ATPs to (public-coin) PQ-ppZKAoK, at the cost of three extra rounds. This compiler, together with [Thm. 1](#), leads to the following result:

Corollary 4. *Assuming the existence of PQ-OWFs, there exists a five-round, public-coin construction of PQ-ppZKAoK (and thus a preprocessing post-quantum strong WIAoK and a preprocessing PQ-WHAoK) where the knowledge extractor is EQPT_c -simulatable and the preprocessing phase has two rounds.*

1.1.4 Application IV: Post-Quantum Non-Malleability in Public-Coin

Non-malleable commitments, introduced by [\[DDN91\]](#), are a type of commitment that protects against *man-in-the-middle attacks*. Specifically, they ensure that an adversary cannot produce a commitment to a related message after seeing a commitment to another message. Non-malleable commitments are a highly versatile and useful primitive. They find applications in constructing other non-malleable primitives like coin-flipping or ZK (e.g., [\[BPS06, PR05\]](#)). Additionally, they are also closely related to information-theoretical objects like non-malleable codes and extractors (e.g., [\[DW09, DPW10, GPR16\]](#)). Most significantly, they serve as a critical ingredient in resolving the exact round complexity of MPC, as well as protecting such protocols against concurrent attacks (e.g., [\[KOS03, PPV08, Wee10, Goy11, GMPP16, BGJ⁺18, CCG⁺19, Can00, Can01, CLOS02, Pas03, PS04, MPR06, BDH⁺17, CLP20\]](#) and more).

Although non-malleable commitments have been extensively studied in the classical setting, *post-quantum* non-malleable commitments (PQ-NMCom) manage to see interesting progress only recently. [\[ABG⁺21\]](#) constructed a constant-round PQ-NMCom w.r.t. the special case of *synchronous schedules*, assuming the super-polynomial quantum hardness of Learning with Errors. In the general (i.e., asynchronous) setting, [\[BLS22\]](#) obtained the first $O(\log^* \lambda)$ -round construction from the minimal assumption of PQ-OWFs. More recently, [\[LPY23\]](#) eventually achieved a constant number of rounds assuming PQ-OWFs. However, all these constructions are private-coin. This is in sharp contrast to the classical setting where public-coin constructions were known, and being utilized in several scenarios, for quite a while (e.g., [\[GGJS12, COSV16, COSV17a, COSV17b\]](#)).

As the last application, we show that our public-coin PQ-ATPs lead to the first public-coin PQ-NMCom. Our starting point is the BLS construction from [\[BLS22\]](#). We observe that their use of private coins arises primarily from three factors: (1) they employ a constant-round PQ-ZK protocol to enforce the committer’s honest behavior in their small tag construction; (2) their tag amplification compiler sets up an FLS-style trapdoor in a private-coin way; (3) they rely on the private-coin ExtCom construction from [\[CCLY22a\]](#).

The main idea underpinning our result is the observation that the PQ-ZK protocol can be replaced by an FLS-style construction, and our new PQ-ATPs can serve as a public-coin “trapdoor.” This PQ-ATP must occur *at the very beginning* of the whole protocol, to enable non-uniform arguments in specific steps of the security proof. We refer to [Sec. 8](#) and [9](#) for further details. Since public-coin PQ-ATPs follow from PQ-OWFs (and so do other components we inherited from the BLS protocol), we obtain the following result.

Corollary 5. *Assuming the existence of PQ-OWFs, there exists a public-coin, $O(\log^* \lambda)$ -round construction of PQ-NMCom.*

1.2 Related Work

As will be detailed in [Sec. 2.1](#), our construction of ATPs is inspired by the prior work of [\[HNY17\]](#), where the authors showed that any average-case hard **NP** problem implies an average-case hard

TFNP problem with non-uniform verifiers. However, we would like to emphasize that [HNY17] is primarily a complexity-theoretic work. To the best of our knowledge, it has not been applied in cryptographic contexts, nor has it been widely recognized within the cryptography community. One of the interesting aspects of our work is the identification of a concrete cryptographic application of this idea. Moreover, our work incorporates additional non-trivial ideas beyond those in [HNY17]. These include the introduction of the notion of preprocessing zero-knowledge, as well as a non-trivial adaptation of the non-malleable commitment scheme from [BLS22] to obtain the first public-coin protocol.

Another related idea appears in [PV20, Proposition 3.2], where the authors use *infinitely often* one-way functions (OWFs) to build a public-coin “2-round puzzle.” We note, however, that the notion of a “2-round puzzle” in [PV20] differs from ours: their construction assumes an honest puzzle generator, making it easier to guarantee the existence of a puzzle solution. Roughly speaking, their approach only requires the (honest) generator to send a randomly sampled (infinitely often) UOWHF along with a collision-target preimage. In contrast, our construction must ensure the existence of a puzzle solution even in the presence of a malicious generator, which necessitates additional technical ideas.

2 Technical Overview

Organization. We begin by introducing our definition and construction of the almost-total puzzle in Sec. 2.1. As noted earlier, this new primitive has multiple applications. In this overview, however, we focus on its use in post-quantum preprocessing zero-knowledge arguments of knowledge, presented in Sec. 2.2, as it most clearly illustrates our core ideas. Additional related applications are briefly discussed in Sec. 2.2.2. Finally, we outline the main ideas underlying our construction of public-coin post-quantum non-malleable commitments in Sec. 2.3.

2.1 Almost-Total Puzzle

A (post-quantum) almost-total puzzle (ATP) is an interactive protocol consisting of PPT ITMs G referred as a puzzle generator and R referred as a puzzle receiver. Upon execution of the protocol $\langle G, R \rangle$, a transcript τ , encoding the puzzle is generated. The generated puzzle must satisfy the following properties:

- **Almost-total:** A solution for puzzle τ must exist with overwhelming probability, even if G is malicious during the interaction.
- **Efficiently-verifiable:** A given solution could be certified efficiently.
- **Hard:** The solution should be hard for any efficient R to solve, except for negligible probability.

An ATP is *public-coin* if G sends only random bits during the interaction.

Construction. Our construction of ATPs makes use of *universal one-way hash functions* (UOWHFs) [NY89], which in turn follow from OWFs, in both the classical [Rom90] and post-quantum [AC12, Section 2.1] settings. In UOWHFs, the adversary $\mathcal{A}$ must fix one preimage before the hash key k is chosen, then select the other preimage after seeing k . The probability that the two preimages collide is negligible for any efficient adversary. Note that k is randomly chosen by the challenger. See the formal definitions in Sec. 3.3.

We now present the construction of a two-round public-coin ATP, which achieves classical (resp. post-quantum) security if the underlying UOWHF is classically (resp. post-quantum) secure:

1. R sends n uniformly random strings $x'_1, \dots, x'_n$ in the domain of UOWHF and send them to G , where n depends on the security parameter λ and UOWHF.
2. G samples a uniformly random string x (in the domain of UOWHF) and a hash key k send them to R .

We say that s is a solution for the ATP if there exists an $i \in [n]$ s.t.

$$(h_k(s) = h_k(x_i)) \wedge (s \neq x_i), \text{ where } x_i := x \oplus x'_i.$$

It is clear from the description that our construction is public-coin and efficiently verifiable. As for hardness, since each x_i is uniformly random due to the randomness of x (when G is honest) and hash key k is sent after R sent $x'_1, \dots, x'_n$, therefore we can simply reduce hardness to the security of UOWHF.

As for the almost-total property, by the compressing property of UOWHF, at least half of the target has a collision. Note that, for each fixed x and k , $x_1, \dots, x_n$ are independently and uniformly random. Thus, for each fixed x and k , at least one of $x_1, \dots, x_n$ has a collision under k except for probability 2^{-n} . By taking the union bound over all possible x 's and k 's, the probability that G can choose x and k in such a way that none of $x_1, \dots, x_n$ has a collision is at most $2^{|x|+|k|} \cdot 2^{-n}$. So if we take n sufficiently large, then the collision probability will be negligible.

Our above approach is inspired by [HNY17], where they showed that any average-case hard **NP** problem implies average-case hard **TFNP** problem with non-uniform verifiers. Intuitively, the non-uniformity is used to hardwire the good choice of $x'_1, \dots, x'_n$ in the above idea. Then, [HNY17] argued how to remove the non-uniformity using an additional derandomization assumption. In our construction, it is the interactions between G and R that remove the non-uniformity. It is worth noting that we could instantiate the above construction with any average-case hard **NP** problem, though this offers no additional benefits since OWFs are already a necessary component for all our applications.

2.2 Post-Quantum Preprocessing ZKAoK

We define a new notion called (post-quantum) preprocessing zero-knowledge arguments of knowledge (ppZKAoK). Essentially, ppZKAoK protocols consist of two stages: a *preprocessing stage* and a *prove stage*. During the preprocessing stage, the prover and verifier interact without any knowledge of the statement that will eventually be proven. The statement will become available at the beginning of the prove stage. This structure extends the concept of *delayed-input* protocols, and its completeness and arguments of knowledge properties are defined similarly to those in delayed-input AoK protocols.⁹

However, the ZK property is defined differently within this preprocessing framework, which we refer to as *preprocessing zero-knowledge* (ppZK)—we allow a two-stage simulator $\mathcal{S}$: an unbounded-time machine $\mathcal{S}_0$ with polynomial-size output and a QPT machine $\mathcal{S}_1$. $\mathcal{S}_0$ takes as input a transcript $\mathbf{pp}$ generated by P and $V^*(\rho)$ in the preprocessing stage, and outputs $\mathcal{S}_0(\mathbf{pp})$. Once the prove stage begins, $\mathcal{S}_1$ takes both $\mathcal{S}_0(\mathbf{pp})$ and the instance x as input to finish the simulation. The ppZK property requires that $\mathcal{S}_1^{V^*(\rho')}(\mathcal{S}_0(\mathbf{pp}), x)$ is indistinguishable from the output of a real execution between $P(w)$ and $V^*(\rho)$ after the prove stage.

This preprocessing zero-knowledge (ppZK) property is sufficient in various settings where the instance is provided to the verifier after preprocessing. Indeed, it implies useful properties such as strong WI and WH, *both with preprocessing* (Def. 24 and 25), as well as *standard* WI (without preprocessing). We refer the reader to Sec. 5.2 for a detailed discussion of the implications of ppZKAoK and its relationship with standard ZKAoK.

⁹ In the post-quantum setting, we require knowledge extraction with EQPT_c simulation

2.2.1 Construction

We show that our public-coin PQ-ATP implies a five-round, *public-coin* ppZKAoK scheme with EQPT_c simulation. Additionally, our construction uses Naor’s commitment scheme and a PQ-WIPoK with EQPT_m -guaranteed-extraction property [LMS22] (which also follow from PQ-OWFs). Let us first recall these notions.

EQPT_m stands for measured-runtime expected quantum polynomial-time. It is a class of quantum simulators that run for an expected polynomial number of steps when executed as follows: at each step, the simulator applies a fixed, constant-size quantum circuit U , followed by a measurement to determine whether to halt or perform another step. In contrast, EQPT_c stands for coherent-runtime expected quantum polynomial-time, which is introduced by [LMS22] as an alternative model for post-quantum simulation. Roughly speaking, an EQPT_c computation can run an EQPT_m procedure coherently, perform some operation on the result, and then apply the inverse of the same procedure. It does not measure when it halts so that computational branches with different runtime can interfere with each other.

We say a PQ-WIPoK has EQPT_m -guaranteed-extraction property if there exist an EQPT_m machine $\mathcal{GE}$ that satisfies the following property: We first run the malicious prover P^* coherently and measure the decision bit of the honest verifier; if the verifier accepts, then we can run $\mathcal{GE}$ to extract a value v . The guaranteed-extraction property ensures that when the verifier accepts, the extracted v is the a valid witness, except for with negligible probability.

With these components, our PQ-ppZKAoK works as follows:

1. In the preprocessing stage, prover P and verifier V run a PQ-ATP, with P acting as the puzzle receiver R and V acting as the puzzle sender G . If the execution completes without aborting, they jointly generate a valid puzzle τ ;
2. In the prove stage, P first sends a commitment c to the witness w of the statement x , using Naor’s commitment. Then, they run a PQ-WIPoK with guaranteed-extraction, where P proves that either it “knows” a solution for τ or c is a commitment of a valid witness $w \in \mathcal{R}_{\mathcal{L}}(x)$.

It is clear from the description that our protocol is complete and public-coin, and we refer the reader to [Sec. 5.3](#) for a discussion about round complexity. We now focus on the preprocessing ZK and adaptive post-quantum AoK with EQPT_c -simulation properties.

For preprocessing ZK: By the almost-total property of ATP, a solution for τ exists with overwhelming probability. Since $\mathcal{S}_0$ is an unbounded-time machine, it can find a solution s for τ with overwhelming probability. In the prove stage, $\mathcal{S}_1$ commits to 0 and uses s as the witness for the PQ-WIPoK. By the (non-uniform) WI of the PQ-WIPoK and the (non-uniform) hiding property of Naor’s commitment scheme, the simulated view is indistinguishable from the verifier’s view in a real execution.

For adaptive AoK with EQPT_c -simulation: Our goal is to construct a post-quantum witness-extended extractor $\mathcal{WE}$ that extracts the witness from any successful prover, while simulating the post-extraction state with EQPT_c -simulation (see [Def. 23](#) for a formal definition). Roughly speaking, machine $\mathcal{WE}$ first *coherently* runs the EQPT_m -guaranteed-extractor $\mathcal{GE}$ we assumed from the PQ-WIPoK. That is, $\mathcal{WE}$ applies the unitary dilation $U_{\mathcal{GE}}$ of $\mathcal{GE}$, measures to obtain the witness, and then applies the inverse $U_{\mathcal{GE}}^\dagger$. Finally, $\mathcal{WE}$ measures and outputs the internal state of the malicious prover. By the guaranteed extraction property, the probability that the verifier accepts the proof but the extracted value is not a witness for the OR-composed language is very small. Also, by the hardness property of ATPs, the probability that the verifier accepts the proof but the extracted value is not a witness of instance x is negligible. Finally, we rely on the statistical binding property

of Naor’s commitment, which essentially guarantees that the value extracted by $\mathcal{WE}$ is fixed with overwhelming probability. Consequently, by the gentle measurement lemma, the extraction process only negligibly alters the internal state of the malicious prover. For further details, we refer the reader to [Sec. 5.5](#).

2.2.2 Other Related Results

We briefly mention other results that are closely related to the above PQ-ppZKAoK.

PQ-ppZKAoK also implies post-quantum strong WIAoK and WHAoK, *both in the preprocessing setting*, as well as standard PQ-WIAoK (i.e., in the “non-preprocessing” setting). For further details, see [Sec. 5.2](#).

The construction of our (classical) dinuZKAoK (as per [Corollary 3](#)) follows the same construction as the above PQ-ppZKAoK, except that the PQ-WIPoK is replaced by a delayed-input WIPoK, which can be built from OWFs [[LS91](#)]. A formal treatment is provided in [Sec. 6](#).

Our PQ-ExtCom (see [Corollary 1](#)) is constructed by adding Naor’s commitment between the preprocessing and prove stages of the PQ-ppZKAoK from [Sec. 2.2](#), with the committer proving that it knows the value inside Naor’s commitment. Roughly, the simulatable extractability follows from the AoK property of PQ-ppZKAoK, and the hiding property follows from the (non-uniform) hiding of Naor’s commitment together with the ppZK property. Further details can be found at [Sec. 7](#). In the classical setting, the same construction also represents the first public-coin, constant-round ExtCom that avoids over-extraction (as per [Corollary 2](#)).

The construction of public-coin PQ-NMCom (as per [Corollary 5](#)) requires additional ideas, which we now discuss in [Sec. 2.3](#).

2.3 Post-Quantum Public-Coin Non-Malleable Commitment

In this section, we provide an overview of our post-quantum public-coin non-malleable commitments. For details, please refer to [Sec. 8](#) and [9](#).

We begin by reviewing the $O(\log^* \lambda)$ -round post-quantum non-malleable commitment (PQ-NMCom) construction by Bitansky, Lin, and Shmueli [[BLS22](#)] (hereafter referred to as the BLS protocol), and explain how we can convert it to a public-coin protocol.

2.3.1 [[BLS22](#)] and Private Coins

Roughly, BLS builds a transformation that modularly obtain non-malleable commitments from any ExtCom. The construction consists of two parts: (1) a base protocol that supports a constant-size tag space, and (2) a tag amplification procedure that will be applied iteratively to support an exponential-size tag space.

Small-Tag Protocol. For the small-tag protocol, their idea is to introduce enough rewinding opportunities for extraction (usually referred to as “safe extraction opportunity”) in the protocol, such that there is always an “ideally scheduled block”. An ideally scheduled block is a sequential block of ExtComs that does not interleave with any messages exchange between $\mathcal{M}$ and the left committer C (who will become the external challenger of the hardness-providing gadget in the security proof); Then, one can extract $\tilde{m}$ from this free slot using known post-quantum rewinding strategies. However, since they are only able to extract from one of the block commitments on the right, they require the sender to give a post-quantum constant-round zero-knowledge argument that all blocks commit to the same value $\tilde{m}$. Given the impossibility of [[CCLY22b](#)], this step will render the whole construction private-coin, even if we could implement the ExtCom in public-coin. Additionally, in

order to use a non-uniform reduction to the hiding property of the ExtCom, they require the underlying ExtCom to have an extra property called *first-message binding*, which roughly means that the first sender message in the protocol fixes the committed value. Unfortunately, our public-coin ExtCom does not satisfy this property. Although BLS also showed that any ExtCom can be easily turned into an ExtCom with first-message binding, such a transformation does not preserve the public-coin property.

Tag Amplification. The tag amplification process converts any non-malleable commitment scheme $\langle C, R \rangle$ with $t \in [3, O(\log(\lambda))]$ bit tags into a non-malleable commitment scheme $\langle \hat{C}, \hat{R} \rangle$ with $T = 2^{t-1}$ bit tags. Roughly, this “tag amplifier” works as follows: To commit a message $\tilde{m}$ under tag $\hat{\mathbf{t}} \in \{0, 1\}^T$, the committer creates T new tags of the form $\mathbf{t}\mathbf{g}_i = (i, \hat{\mathbf{t}}[i]) \in \{0, 1\}^t$ corresponding to the base scheme (where $\hat{\mathbf{t}}[i]$ is the i -th bit of $\hat{\mathbf{t}}$). The committer sends T commitments of $\tilde{m}$ in parallel under each one of the tags $\mathbf{t}\mathbf{g}_i$, using the base (i.e., small-tag) scheme $\langle C, R \rangle$. To ensure all commitments commit to the same message, [BLS22] employs the FLS paradigm. However, the way they set up the trapdoor introduces more private-coin usage to the resulting protocol.

2.3.2 Adapting the BLS Protocol to a Public-Coin Scheme

Our construction builds on the [BLS22] protocol but with a focus on achieving public-coin property. For the small-tag protocol, we replace the PQ-ZK argument used in BLS with a WIAoK, and use our post-quantum almost-total puzzle to set up a trapdoor at the beginning of the protocol. We further modify the definition of an ideally scheduled block, allowing the right-side ideally scheduled block to interleave with the left-side almost-total puzzle but not with the left-side WIAoK. We prove that such an adjustment ensures the existence of an ideally scheduled block on the right, using a similar combinatorial argument from BLS. Moreover, we position our almost-total puzzle (as the trapdoor set-up for our WIAoK) at the start of the protocol; we show that such a strategy will eventually enables certain non-uniform arguments in various steps of the security proof to go through, which were crucial in the original [BLS22] paradigm.

Additionally, BLS relies on the first-message binding property of the underlying ExtCom in certain steps of the security proof. We demonstrate that our ExtCom construction, which incorporates a *rewinding-safe binding* property, already suffice. Roughly, rewinding-safe binding ensures that the first sender message that binds the commitment value occurs prior to the ExtCom’s extraction mechanism.

3 Preliminaries

3.1 Basic Notations

Let $\lambda \in \mathbb{N}$ denote security parameter. For a positive integer n , let $[n]$ denote the set $\{1, 2, \dots, n\}$. For a finite set $\mathcal{X}$, $x \leftarrow \mathcal{X}$ means that x is uniformly chosen from $\mathcal{X}$.

A function $f : \mathbb{N} \rightarrow [0, 1]$ is said to be *negligible* if for all polynomial p and sufficiently large $\lambda \in \mathbb{N}$, we have $f(\lambda) < 1/p(\lambda)$; it is said to be *overwhelming* if $1 - f$ is negligible, and said to be *noticeable* if there is a polynomial p such that $f(\lambda) \geq 1/p(\lambda)$ for sufficiently large $\lambda \in \mathbb{N}$. We denote by **poly** an unspecified polynomial and by **negl** an unspecified negligible function.

Honest (classical) parties are modeled as interactive Turing machines (ITMs). We use PPT and QPT to denote (classical) probabilistic polynomial time and quantum polynomial time, respectively. For a classical probabilistic or quantum algorithm $\mathcal{A}$, $y \leftarrow \mathcal{A}(x)$ means that $\mathcal{A}$ is run on input x and outputs y . An adversary (or malicious party) is modeled as a non-uniform PPT (or QPT in the case of post-quantum security) algorithm. When we consider a non-uniform QPT adversary, we often

specify it by a sequence of polynomial-size quantum circuits with quantum advice $\{\mathcal{A}_\lambda, \rho_\lambda\}_{\lambda \in \mathbb{N}}$. In an execution with the security parameter λ , $\mathcal{A}$ runs $\mathcal{A}_\lambda$ taking ρ_λ as the advice. For simplicity, we often omit the index λ and just write $\mathcal{A}(\rho)$ to mean a non-uniform QPT algorithm specified by $\{\mathcal{A}_\lambda, \rho_\lambda\}_{\lambda \in \mathbb{N}}$.

For an **NP** language $\mathcal{L}$ and a true statement in this language $x \in \mathcal{L}$, we use $\mathcal{R}_\mathcal{L}(x)$ ($\mathcal{R}$ stands for “relation”) to denote the set of all witnesses for x . We will refer to the OR-composition of **NP** languages, which are defined in [Def. 1](#). The following definitions are based on those in [\[LPY23\]](#)

Definition 1 (OR-Composition of NP Languages). *Let $\mathcal{L}_1$ and $\mathcal{L}_2$ be two NP languages. The OR-composition of them (dubbed $\mathcal{L}_1 \vee \mathcal{L}_2$) is the new NP language defined as follows:*

$$\mathcal{L}_1 \vee \mathcal{L}_2 := \{(x_1, x_2) \mid x_1 \in \mathcal{L}_1 \vee x_2 \in \mathcal{L}_2\}.$$

Notations for Indistinguishability. We may consider random variables over bit strings or over quantum states. This will be clear from the context. We use the same notations for classical and quantum computational indistinguishability, but there should be no fear of confusion; It means computational indistinguishability against PPT (resp. QPT) distinguishers whenever we consider classical (resp. post-quantum) security. For ensembles of random variables $\mathcal{X} = \{X_i\}_{\lambda \in \mathbb{N}, i \in I_\lambda}$ and $\mathcal{Y} = \{Y_i\}_{\lambda \in \mathbb{N}, i \in I_\lambda}$ over the same set of indices $I = \bigcup_{\lambda \in \mathbb{N}} I_\lambda$ and a function δ , we use $\mathcal{X} \stackrel{c}{\approx}_\delta \mathcal{Y}$ to mean that for any *non-uniform* PPT (resp. QPT) algorithm $\mathcal{A}$, there exists a negligible function $\text{negl}(\cdot)$ such that for all $\lambda \in \mathbb{N}$, $i \in I_\lambda$, we have

$$|\Pr[\mathcal{A}(X_i)] - \Pr[\mathcal{A}(Y_i)]| \leq \delta(\lambda) + \text{negl}(\lambda).$$

We say that $\mathcal{X}$ and $\mathcal{Y}$ are δ -computationally indistinguishable if the above holds. In particular, when the above holds for $\delta = 0$, we say that $\mathcal{X}$ and $\mathcal{Y}$ are computationally indistinguishable, and simply write $\mathcal{X} \stackrel{c}{\approx} \mathcal{Y}$.

Similarly, we use $\mathcal{X} \stackrel{s}{\approx}_\delta \mathcal{Y}$ to mean that for any unbounded time algorithm $\mathcal{A}$, there exists a negligible function $\text{negl}(\cdot)$ such that for all $\lambda \in \mathbb{N}$, $i \in I_\lambda$, we have

$$|\Pr[\mathcal{A}(X_i)] - \Pr[\mathcal{A}(Y_i)]| \leq \delta(\lambda) + \text{negl}(\lambda).$$

In particular, when the above hold for $\delta = 0$, we say that $\mathcal{X}$ and $\mathcal{Y}$ are statistically indistinguishable, and simply write $\mathcal{X} \stackrel{s}{\approx} \mathcal{Y}$. Moreover, we write $\mathcal{X} \equiv \mathcal{Y}$ to mean that X_i and Y_i are distributed identically for all $i \in I$.

When we consider an ensemble $\mathcal{X}$ that is only indexed by λ (i.e., $I_\lambda = \{\lambda\}$), we write $\mathcal{X} = \{X_\lambda\}_\lambda$ for simplicity.

3.2 Expected Quantum Polynomial-Time Computations

We briefly review two notions of expected quantum polynomial-time: (1) measured-runtime expected quantum polynomial-time (dubbed EQPT_m), and (2) the coherent-runtime expected quantum polynomial-time (dubbed EQPT_c) introduced in [\[LMS22\]](#).

The definition of EQPT_m is rather intuitive. Suppose that we run a quantum Turing machine where we measure if it is in a “halt state” after each application of the transition unitary. We say that it is EQPT_m if the expected number of steps before halting is polynomial in the input length.

The definition of EQPT_c is more complicated. Roughly speaking, an EQPT_c computation does not measure when it halts so that computational branches with different runtime can interfere with each other. For the purpose of this paper, there is no need to recall the definition of EQPT_c in its full detail (for which we refer to [\[LMS22\]](#)). We only need a special case of what can be done in EQPT_c as phrased in the following fact:

Fact 2. Let Φ be a EQPT_m quantum operation and U_Φ be its unitary dilation. That is, U_Φ is a unitary on the input register of Φ and an ancilla register such that Φ is equivalent to appending the ancilla register initialized to $|0\dots 0\rangle$, applying U_Φ , and discarding the ancilla. Let $M^{U_\Phi, U_\Phi^\dagger}$ be a QPT oracle machine that makes one query to each of unitary oracles U_Φ and $U_\Phi^\dagger$ where the query to U_Φ is made before that to $U_\Phi^\dagger$ and acts on the ancilla register of Φ only through access to U_Φ and $U_\Phi^\dagger$. Then an execution of $M^{U_\Phi, U_\Phi^\dagger}$ with the ancilla of Φ initialized to $|0\dots 0\rangle$ can be done in EQPT_c .

3.3 Universal One-Way Hash Function

We recall the definition of (classical and post-quantum) universal one-way hash functions (UOWHF). The following definitions are based on those in [KK05]

Definition 2 (Classical Universal One-Way Hash Function). Let $\ell_{\text{in}}(\cdot)$, $\ell_{\text{out}}(\cdot)$, and $\ell_{\text{key}}(\cdot)$ be polynomials. A collection of function families $\mathcal{H} = \{H_\lambda\}_{\lambda \in \mathbb{N}}$, where each H_λ is a function family $H_\lambda = \{h_k : \{0, 1\}^{\ell_{\text{in}}(\lambda)} \rightarrow \{0, 1\}^{\ell_{\text{out}}(\lambda)}\}_{k \in \{0, 1\}^{\ell_{\text{key}}(\lambda)}}$ is a universal one-way hash family (UOWHF) if:

- **Efficiency:** There is a polynomial $\text{poly}(\cdot)$ such that for all $\lambda \in \mathbb{N}$, $k \in \{0, 1\}^{\ell_{\text{key}}(\lambda)}$, and $x \in \{0, 1\}^{\ell_{\text{in}}(\lambda)}$, the function $h_k(x)$ can be computed in $\text{poly}(\lambda)$ time.
- **Compression:** For all $\lambda \in \mathbb{N}$, we have $\ell_{\text{out}}(\lambda) < \ell_{\text{in}}(\lambda)$.
- **Universal One-Wayness:** For all (non-uniform) PPT algorithms $\mathcal{A}$, it holds that

$$\Pr \left[(x \neq x') \wedge (h_k(x) = h_k(x')) : \begin{array}{l} x \leftarrow \mathcal{A}(1^\lambda); k \xleftarrow{\$} \{0, 1\}^{\ell_{\text{key}}(\lambda)} \\ x' \leftarrow \mathcal{A}(1^\lambda, k, x) \end{array} \right] = \text{negl}(\lambda).$$

Definition 3 (Post-Quantum Universal One-Way Hash Function). Using the same notation as in Def. 2. A collection of function families $\mathcal{H} = \{H_\lambda\}_{\lambda \in \mathbb{N}}$ is post-quantum universal one-way hash family if it satisfies the same **Efficiency** and **Compression** requirements as in Def. 2, and the **Post-Quantum Universal One-Wayness** formalized as follows:

- **Post-Quantum Universal One-Wayness:** For all non-uniform QPT algorithms $\mathcal{A} = \{(\mathcal{A}_\lambda^1, \mathcal{A}_\lambda^2, \rho_\lambda)\}_{\lambda \in \mathbb{N}}$, it holds that

$$\Pr \left[(x \neq x') \wedge (h_k(x) = h_k(x')) : \begin{array}{l} (x, \rho'_\lambda) \leftarrow \mathcal{A}_\lambda^1(\rho_\lambda); k \xleftarrow{\$} \{0, 1\}^{\ell_{\text{key}}(\lambda)} \\ x' \leftarrow \mathcal{A}_\lambda^2(k, x, \rho'_\lambda) \end{array} \right] = \text{negl}(\lambda).$$

Remark 2 (Public-Coin). Note that in Def. 2 and 3, the “key generation” procedure is simply sampling a k uniformly at random from $\{0, 1\}^{\ell_{\text{key}}(\lambda)}$. This implies that the defined UOWHF family is *public-coin*, i.e., the family remains collision-resistant even if the randomness used to generate the key is known to the adversary. This property is crucial as our results (to appear) focus on the public-coin regime.

3.4 Interactive Proofs and Arguments

We define interactive proofs and arguments. An interactive protocol consists of PPT ITMs P referred to as a prover and V referred to as a verifier. We denote by $\langle P(x_P), V(x_V) \rangle(x)$ an execution of the protocol where x is a common input, x_P is P ’s private input, and x_V is V ’s private input. We denote by $\text{OUT}_V \langle P(x_P), V(x_V) \rangle(x)$ the final output of V in the execution. An honest verifier’s output is $\top$ indicating acceptance or $\perp$ indicating rejection, and a quantum malicious verifier’s output may be an arbitrary quantum state.

Definition 4 (PQ-IP and PQ-IA). A post-quantum interactive proof (PQ-IP) (resp. argument (PQ-IA)) for an NP language $\mathcal{L}$ is an interactive protocol $\langle P, V \rangle$ satisfying the following:

- **Completeness:** There exists a negligible function negl such that for any $\lambda \in \mathbb{N}$, $x \in L \cap \{0, 1\}^\lambda$, and $w \in \mathcal{R}_{\mathcal{L}}(x)$, it holds that

$$\Pr[\text{OUT}_V \langle P(w), V \rangle(x) = \top] = 1 - \text{negl}(\lambda).$$

- **Statistical (resp. Computational) Soundness:** For any unbounded-time (resp. non-uniform QPT) machine $\{P_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$, there exists a negligible function negl such that for any $\lambda \in \mathbb{N}$ and any $x \in \{0, 1\}^\lambda \setminus \mathcal{L}$, it holds that

$$\Pr[\text{OUT}_V \langle P_\lambda^*(\rho_\lambda), V \rangle(x) = \top] \leq \text{negl}(\lambda).$$

Definition 5 (PQ-WIP and PQ-WIA). A post-quantum witness indistinguishable proof (PQ-WIP) (resp. argument (PQ-WIA)) is a PQ-IP (resp. PQ-IA) that additionally satisfies the following:

- **Witness Indistinguishability:** For any non-uniform QPT machine $\{V_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$, we have

$$\{\text{OUT}_{V_\lambda^*} \langle P(w_0), V_\lambda^*(\rho_\lambda) \rangle(x)\}_{\lambda, x, w_0, w_1} \stackrel{c}{\approx} \{\text{OUT}_{V_\lambda^*} \langle P(w_1), V_\lambda^*(\rho_\lambda) \rangle(x)\}_{\lambda, x, w_0, w_1},$$

where $\lambda \in \mathbb{N}$, $x \in \mathcal{L} \cap \{0, 1\}^\lambda$, and $w_0, w_1 \in \mathcal{R}_{\mathcal{L}}(x)$.

Delayed-Input Interactive Argument Next, let us recall the definition of delayed-input interactive arguments [JKKR17], and the definition of strong witness indistinguishability (strong WI) and its extension to the delayed-input setting.

Definition 6 (Delayed-Input Interactive Argument). A classical n -round interactive protocol for an NP language $\mathcal{L}$ is called delayed-input if it satisfies the following:

- **Completeness:** There exists a negligible function negl such that for any $\lambda \in \mathbb{N}$, $x \in L \cap \{0, 1\}^\lambda$, and $w \in \mathcal{R}_{\mathcal{L}}(x)$, it holds that

$$\Pr[\text{OUT}_V \langle P(w), V \rangle(x) = \top] = 1 - \text{negl}(\lambda).$$

- **Adaptive Soundness:** For every $z \in \{0, 1\}^*$, every PPT machine P^* that chooses $x \in \{0, 1\}^\lambda \setminus \mathcal{L}$ adaptively, depending upon the first $n - 1$ rounds, there exists a negligible function negl such that

$$\Pr[\text{OUT}_V \langle P^*, V \rangle(x) = \top] \leq \text{negl}(\lambda).$$

Definition 7 ((Delayed-Input) Strong WI). An interactive argument (resp., a delayed-input interactive argument) for an NP language $\mathcal{L}$ is called strongly witness indistinguishable (resp., delayed-input strongly witness indistinguishable) if the following holds: satisfies the following:

For every $\{\mathcal{X}_\lambda^0, \mathcal{W}_\lambda^0\}_{\lambda \in \mathbb{N}}$, $\{\mathcal{X}_\lambda^1, \mathcal{W}_\lambda^1\}_{\lambda \in \mathbb{N}}$ and $\{z_\lambda\}_{\lambda \in \mathbb{N}}$ where each $\mathcal{X}_\lambda^b, \mathcal{W}_\lambda^b$ is a joint distribution that ranges over $\mathcal{R}_{\mathcal{L}} \cap (\{0, 1\}^\lambda \times \{0, 1\}^*)$ and each z_λ is a string in $\{0, 1\}^*$, if it holds

$$\{\mathcal{X}_\lambda^0\}_{\lambda \in \mathbb{N}} \stackrel{c}{\approx} \{\mathcal{X}_\lambda^1\}_{\lambda \in \mathbb{N}}$$

then for every PPT verifier (resp. PPT delayed-input verifier) V^* , there exists a negligible function negl such that for any $\lambda \in \mathbb{N}$, $x \in L \cap \{0, 1\}^\lambda$, it holds that

$$\begin{aligned} & \Pr[\text{OUT}_V \langle P(w), V^*(z_\lambda) \rangle(x) = \top : (x, w) \leftarrow (\mathcal{X}_\lambda^0, \mathcal{W}_\lambda^0)] \\ & \leq \Pr[\text{OUT}_V \langle P(w), V^*(z_\lambda) \rangle(x) = \top : (x, w) \leftarrow (\mathcal{X}_\lambda^1, \mathcal{W}_\lambda^1)] + \text{negl}(\lambda) \end{aligned}$$

Definition 8 (Delayed-Input Witness Hiding). A delayed-input interactive argument $\langle P, V \rangle$ for an NP language $\mathcal{L}$ satisfies delayed-input witness hiding if it satisfies the following:

- **Delayed-Input Witness Hiding:** For any delayed-input non-uniform PPT machine V^* , $\{z_\lambda\}_{\lambda \in \mathbb{N}}$ and $\{a_\lambda\}_{\lambda \in \mathbb{N}}$, there is a non-uniform PPT machine M such that for any family of distributions $\{(\mathcal{X}_\lambda, \mathcal{W}_\lambda)\}_{\lambda \in \mathbb{N}}$ where $(\mathcal{X}_\lambda, \mathcal{W}_\lambda)$ is a joint distribution supported on $\mathcal{R}_\mathcal{L} \cap (\{0, 1\}^\lambda \times \{0, 1\}^*)$ each z_λ and a_λ are strings in $\{0, 1\}^*$, it holds that

$$\begin{aligned} & \Pr[\text{OUT}_{V^*} \langle P(w), V^*(z_\lambda) \rangle(x) \in \mathcal{R}_\mathcal{L}(x) : (x, w) \leftarrow (\mathcal{X}_\lambda, \mathcal{W}_\lambda)] \\ & \leq \Pr[M(a_\lambda, x) \in \mathcal{R}_\mathcal{L}(x) : (x, w) \leftarrow (\mathcal{X}_\lambda, \mathcal{W}_\lambda)] + \text{negl}(\lambda) \end{aligned}$$

Post-Quantum Witness Indistinguishable Argument of Knowledge (PQ-WIAoK) Next, let us review the definition of PQ-WIAoK. The following definition is a combination of [LMS22, Definition 1.5] and [LPY23, Definition 14].

Definition 9 (PQ-WIAoK with EQPT_c-Simulation). A post-quantum witness indistinguishable argument of knowledge (PQ-WIAoK) is a PQ-WIA that additionally satisfies the following:

- **AoK with EQPT_c-Simulation** There exists an EQPT_c oracle machine $\mathcal{WE}$, called the witness-extended emulator, such that for any non-uniform QPT machine $\{P_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$ the following conditions hold:

$$1. \{(\text{ST}'_{P^*}, b') : (\text{ST}'_{P^*}, b', w') \leftarrow \mathcal{WE}^{P_\lambda^*(\rho_\lambda)}(x)\}_{\lambda, x} \stackrel{c}{\approx} \{(\text{ST}_{P^*}, b) : (\text{ST}_{P^*}, b) \leftarrow \langle P_\lambda^*(\rho_\lambda), V(x) \rangle\}_{\lambda, x} \text{ where } \lambda \in \mathbb{N} \text{ and } x \in \{0, 1\}^\lambda,$$

$$2. \Pr[b' = \top \wedge w' \notin \mathcal{R}_\mathcal{L}(x) : (\text{ST}'_{P^*}, b', w') \leftarrow \mathcal{WE}^{P_\lambda^*(\rho_\lambda)}(x)] = 0 \text{ for all } x,^{10}$$

where $(\text{ST}_{P^*}, b) \leftarrow \langle P_\lambda^*(\rho_\lambda^*), V(x) \rangle$ means that ST_{P^*} is the final state of P_λ^* and b is the decision output by V .

Definition 10 (PQ-WIAoK without post-extraction simulation). A post-quantum witness indistinguishable argument of knowledge (PQ-WIAoK) without post-extraction simulation is a PQ-WIA that additionally satisfies the following:

- **Knowledge Soundness** There is an oracle-aided QPT algorithm $\mathcal{K}$, a polynomial poly , a negligible function negl , and a constant $d \in \mathbb{N}$ such that for any QPT malicious prover $P^* = \{P_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$, $\lambda \in \mathbb{N}$ and $x \in \{0, 1\}^\lambda$, we have

$$\Pr[w' \in \mathcal{R}_\mathcal{L}(x) : w' \leftarrow \mathcal{K}^{P_\lambda^*(\rho_\lambda)}(x)] \geq \frac{1}{\text{poly}(\lambda)} \cdot \Pr[\text{OUT}_V \langle P_\lambda^*(\rho_\lambda), V \rangle(x) = \top]^d - \text{negl}(\lambda).$$

Proof of Knowledge with Guaranteed Extraction.

Following [LMS22], we define another type of witness-extractability called guaranteed extraction. The notion is defined only for four-round public-coin proofs (i.e., Σ -protocols).¹¹

Definition 11 (PQ-WIPoK with EQPT_m-Guaranteed-Extraction [LMS22]). We say that a four-round public-coin PQ-WIP is a PQ-WIPoK with EQPT_m-guaranteed-extraction if there is an EQPT_m oracle machine $\mathcal{GE}$, called the guaranteed extractor, that satisfies the following: For an unbounded-time machine $\{P_\lambda^*, \rho_\lambda\}_\lambda$, let $\text{Exp}_{\mathcal{GE}}(\lambda)$ be the following experiment:

¹⁰ Though the original definition in [LMS22] allows the probability to be $\text{negl}(|x|)$, we can assume it to be 0 without loss of generality. If the probability is $\text{negl}(|x|)$, we can modify $\mathcal{WE}$ to output $b' = \perp$ whenever $w' \notin \mathcal{R}_\mathcal{L}(x)$. This only negligibly affects the distribution in the LHS of Condition 1, which can be absorbed into $\stackrel{c}{\approx}$.

¹¹ In [LMS22], they also define it for arguments, but we only need the case of proofs.

1. On security parameter 1^λ , generate the first-round message v of the verifier.
2. Run P_λ^* on the initial state ρ_λ and the first-round message v to generate an instance x and the second-round message a .¹²
3. Generate a uniform superposition $\sum_{r \in R} |r\rangle$ over the possible values of the third-round message (which is public-coin), and coherently run P_λ^* to generate the fourth-round message in superposition.
4. Measure the verifier's decision on the transcript formed by v , a , and the values in the third-round and fourth-round message registers w.r.t. the instance x (without measuring the values of the third-round and fourth messages themselves). Let b be the measurement outcome.
5. If $b = \perp$, output $(x, b = \perp, w = \perp)$.
6. If $b = \top$, run $\mathcal{GE}^{P_\lambda^*}(\rho'_\lambda)(v, a, x)$ to obtain w where ρ'_λ is the internal state of P_λ^* right after [Step 4](#). Output $(x, b = \top, w)$.

Then it holds that

$$\Pr[b = \top \wedge w \notin \mathcal{R}_\mathcal{L}(x) : (x, b, w) \leftarrow \text{Exp}_{\mathcal{GE}}(\lambda)] \leq \text{negl}(\lambda).$$

Theorem 3 ([LMS22]). *Assuming the existence of PQ-OWFs, there exist four-round public-coin PQ-WIPoKs with EQPT_m-guaranteed-extraction.*

3.5 Commitment Schemes

We define post-quantum secure commitments. The following definitions are taken verbatim from [\[LPY23\]](#).

Definition 12 (Commitment). *A commitment scheme is a pair of PPT ITMs $\langle C, R \rangle$ that satisfies the following. Let $m \in \{0, 1\}^{\ell(\lambda)}$ (where $\ell(\cdot)$ is some polynomial) is a message that C wants to commit to. The protocol consists of the following stages:*

- **Commit Stage:** $C(m)$ and R interact with each other to generate a transcript τ , a decommitment decom as C 's private output, and a decision value $b_{\text{com}} \in \{\top, \perp\}$ as R 's private output. We denote this execution by $(\tau, \text{decom}, b_{\text{com}}) \leftarrow \langle C(m), R \rangle(1^\lambda)$. We say C is non-aborting if $b_{\text{com}} = \top$.
- **Decommit Stage:** C sends the committed message m and the decommitment decom to R , and R outputs decision $b_{\text{dec}} \in \{\top, \perp\}$. We denote this execution by $b_{\text{dec}} \leftarrow \text{Verify}(\tau, m, \text{decom})$.¹³ W.l.o.g., we assume that whenever R rejects at the end of the commit stage (i.e., $b_{\text{com}} = \perp$), R will reject at the end of decommit stage (i.e., $b_{\text{dec}} = \perp$). (Note that w.l.o.g., one can think that b_{com} is included in τ , because we can always modify the protocol to ask R to send b_{com} as the last-round message of the commit stage.)

The scheme satisfies the following completeness requirement:

1. **Completeness.** For any polynomial $\ell : \mathbb{N} \rightarrow \mathbb{N}$ and any $m \in \{0, 1\}^{\ell(\lambda)}$, it holds that

$$\Pr \left[b_{\text{com}} = b_{\text{dec}} = \top : \begin{array}{l} (\tau, \text{decom}, b_{\text{com}}) \leftarrow \langle C(m), R \rangle(1^\lambda) \\ b_{\text{dec}} \leftarrow \text{Verify}(\text{com}, m, \text{decom}) \end{array} \right] = 1.$$

¹² Similarly to the definition in [\[LMS22\]](#), we allow the cheating prover to adaptively choose the instance x after receiving the first-round message.

¹³ We assume that there is no state information for R kept from the commit stage.

Definition 13 (Computationally Hiding). A commitment scheme $\langle C, R \rangle$ is post-quantum computationally hiding if for any non-uniform QPT receiver $R^* = \{(R_\lambda^*, \rho_\lambda)\}_{\lambda \in \mathbb{N}}$ and any polynomial $\ell : \mathbb{N} \rightarrow \mathbb{N}$, $m_0, m_1 \in \{0, 1\}^{\ell(\lambda)}$, the following holds:

$$\{\text{OUT}_{R_\lambda^*} \langle C(m_0), R_\lambda^*(\rho_\lambda) \rangle (1^\lambda)\}_{\lambda, m_0, m_1} \stackrel{c}{\approx} \{\text{OUT}_{R_\lambda^*} \langle C(m_1), R_\lambda^*(\rho_\lambda) \rangle (1^\lambda)\}_{\lambda, m_0, m_1}$$

where $\text{OUT}_{R_\lambda^*} \langle C(m_b), R_\lambda^*(\rho_\lambda) \rangle (1^\lambda)$ ($b \in \{0, 1\}$) denotes the output of R_λ^* at the end of the commit stage, which can be quantum.

Definition 14 (Statistically Binding). A commitment scheme $\langle C, R \rangle$ is statistically binding if for any unbounded-time committer C^* , the following holds:

$$\Pr \left[\exists m_0, m_1, \text{decom}_0, \text{decom}_1, \text{ s.t. } m_0 \neq m_1 \wedge \text{Verify}(\tau, m_0, \text{decom}_0) = \text{Verify}(\tau, m_1, \text{decom}_1) = \top : (\tau, \text{decom}, b_{\text{com}}) \leftarrow \langle C^*, R \rangle (1^\lambda) \right] = \text{negl}(\lambda).$$

Definition 15 (Committed Values). For a commitment scheme $\langle C, R \rangle$, we define the value function as follows:

$$\text{val}(\tau) := \begin{cases} m & \text{if } \exists \text{ unique } m \text{ s.t. } \exists \text{ decom, } \text{Verify}(\tau, m, \text{decom}) = 1 \\ \perp & \text{otherwise} \end{cases},$$

where τ and decom are defined in the commit stage in [Def. 12](#).

3.6 Non-Malleable Commitments

Post-Quantum Non-Malleable Commitments. This definition follows the formalization in [\[LPV08, GPR16\]](#). We consider a QPT man-in-the-middle adversary $\mathcal{M}$ interacting with a committer C in the *left*, and a receiver R in the *right*. We denote the relevant entities used in the right interaction as the “tilde’d” version of the corresponding entities on the left. In particular, suppose that C commits to m in the left interaction, and $\mathcal{M}$ commits to $\tilde{m}$ on the right, i.e., we set $\tilde{m} = \text{val}(\tilde{\tau})$ where $\tilde{\tau}$ is the transcript of the right session. Let $\text{mim}_{\langle C, R \rangle}^{\mathcal{M}}(\lambda, m, \rho)$ denote the random variable that is the pair $(\text{OUT}_{\mathcal{M}}, \tilde{m})$, consisting of $\mathcal{M}$ ’s output (could be a quantum state) as well as the value committed to by $\mathcal{M}$ on the right (assuming C commits to m on the left), where ρ is $\mathcal{M}$ ’s non-uniform quantum advice. We use a *tag-based* (or “identity-based”) specification, and ensure that $\mathcal{M}$ uses a distinct tag $\tilde{t}$ on the right from the tag t it uses on the left. This is done by stipulating that $\text{mim}_{\langle C, R \rangle}^{\mathcal{M}}(\lambda, m, \rho)$ outputs a special value $\perp_{\text{tag}}$ when $\mathcal{M}$ uses the same tag in both the left and right executions. The reasoning is that this corresponds to the uninteresting case when $\mathcal{M}$ is simply acting as a channel, forwarding messages from C on the left to R on the right and vice versa.

Definition 16 (Post-Quantum Non-Malleable Commitments). A commitment scheme $\langle C, R \rangle$ is said to be post-quantum non-malleable if for every (non-uniform) QPT man-in-the-middle adversary $\mathcal{M} = \{\mathcal{M}_\lambda, \rho_\lambda\}_{\lambda \in \mathbb{N}}$ and every polynomial $\ell : \mathbb{N} \rightarrow \mathbb{N}$, it holds that

$$\{\text{mim}_{\langle C, R \rangle}^{\mathcal{M}_\lambda}(\lambda, m_0, \rho_\lambda)\}_{\lambda \in \mathbb{N}, m_0, m_1 \in \{0, 1\}^{\ell(\lambda)}} \stackrel{c}{\approx} \{\text{mim}_{\langle C, R \rangle}^{\mathcal{M}_\lambda}(\lambda, m_1, \rho_\lambda)\}_{\lambda \in \mathbb{N}, m_0, m_1 \in \{0, 1\}^{\ell(\lambda)}}$$

where “ $\stackrel{c}{\approx}$ ” refers to computational indistinguishability against QPT distinguishers.

Remark 3 (On Entangled Auxiliary Information.) The above definition does not consider entanglement between $\mathcal{M}$'s auxiliary input and distinguisher's auxiliary input. However, [BLS22, Claim 3.1] shows that the above definition implies the version that considers such entanglement.

Synchronous Adversaries: This notion refers to man-in-the-middle adversaries who upon receiving a message in the left (reps. right) session, immediately respond with the corresponding message in the right (resp. left) session. An adversary is said to be *asynchronous* if it is not synchronous.

3.7 Extractable Commitments

3.7.1 Committed Value Oracle

In order to define post-quantum extractable commitment, we first need to define something called committed value oracle. The following definitions are taken from [BLS22] with some notational adaptations.

Definition 17. Let $\langle C, R \rangle$ be a (possibly tag-based) commitment scheme. A sequential committed-value oracle $O^\infty[\langle C, R \rangle]$ of $\langle C, R \rangle$ acts as follows in interaction with a committer C^* : it interacts with C^* in many sequential sessions; in each session,

- it participates with C^* in the commit phase of $\langle C, R \rangle$ as the honest receiver R (using a tag chosen adaptively by C^*), obtaining a transcript τ , **and**
- if C^* is non-aborting in the commit phase and sends request **break**, it returns $\text{val}(\tau)$.

The single-session oracle $O^1[\langle C, R \rangle]$ is similar to $O^\infty[\langle C, R \rangle]$, except that it interacts with the adversary in a single session. When the commitment scheme is clear from the context, we write O^∞ (or O^1) for simplicity.

Post-Quantum (Sequentially) Extractable Commitments. A commitment scheme is extractable if there exists an efficient extractor such that, the committed value can be extracted. In the classical setting, constant-round constructions of extractable commitments are known from OWFs [PW09]. In the post-quantum setting, we often need an extractor that (almost) does not disturb the (potentially malicious) committer's state during the extraction. However, in the EQPT_m model, it is not known that such a post-quantum extractable commitments exist from (polynomially hard) post-quantum OWFs. In recent work of [LMS22], they showed that there exists a constant-round EQPT_c -extractable commitment scheme with EQPT_c state-preserving extraction, assuming post-quantum one-way functions. We therefore provide definition for ExtCom with EQPT_c -simulation here.

Definition 18 (PQ-ExtCom with EQPT_c -Simulation). A commitment scheme $\langle C, R \rangle$ is said to be a post-quantum extractable commitment with EQPT_c -simulation if it satisfies the following in addition to post-quantum computational hiding and statistical binding.

- **Extraction with EQPT_c -Simulation** There exists an EQPT_c oracle machine $\mathcal{SE}_c$ such that for any non-uniform QPT committer $\{C_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$,

$$\{\mathcal{SE}_c(1^\lambda, C_\lambda^*, \rho_\lambda)\}_{\lambda \in \mathbb{N}} \stackrel{c}{\approx} \{\text{OUT}_{C_\lambda^*}(C_\lambda^{*O^\infty}(\rho_\lambda))\}_{\lambda \in \mathbb{N}}$$

where “ $\stackrel{c}{\approx}$ ” refers to computational indistinguishability against QPT distinguishers.

We also give definition for another variant of post-quantum extractable commitment called ε -simulatable commitment. It was first proposed by [CCLY22a] which allows an (arbitrarily small) noticeable simulation error in extractability. Their definition of post-quantum ε -simulatable commitment is given below.

Definition 19 (Post-Quantum ε -Simulatable Extractable Commitment). *A commitment scheme $\langle C, R \rangle$ is said to be a post-quantum ε -simulatable extractable commitment if it satisfies the following in addition to post-quantum computational hiding and statistical binding. There exists a QPT algorithm $\mathcal{SE}_\varepsilon$, called the ε -simulatable extractor, such that for any noticeable $\varepsilon(\lambda)$ and any non-uniform QPT committer $\{C_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$,*

$$\{\mathcal{SE}(1^\lambda, 1^{\varepsilon^{-1}}, C_\lambda^*, \rho_\lambda)\}_{\lambda \in \mathbb{N}} \stackrel{c}{\approx}_\varepsilon \{C_\lambda^{*O^\infty}(\rho_\lambda)\}_{\lambda \in \mathbb{N}},$$

where “ $\stackrel{c}{\approx}_\varepsilon$ ” refers to ε -close computational indistinguishability against QPT distinguishers.

Lemma 1 ([LMS22, Claim 9.4.]). *Post-quantum extractable commitments with EQPT_c -simulation implies ε -simulatable extractable commitments.*

Remark 4. In both Def. 18 and Def. 19, we use O^∞ instead of O^1 . This is due to the fact that the O^∞ and O^1 versions of any extractable commitments are equivalent, shown in [BLS22, Lemma 3.2]. The former version is also called *sequentially extractable* commitment in their definition.

3.7.2 r -Robustness

We also define the *robust* extractability following [BLS22]. Roughly speaking, r -robust extractability means that extractability holds even if the adversary interacts with an external r -round interactive machine that cannot be rewound by the extractor. The formal definition is given below.

Definition 20 (Post-Quantum r -Robust Extractability). *A commitment scheme $\langle C, R \rangle$ is said to be a post-quantum r -robust extractable if it satisfies the following in addition to post-quantum computational hiding and statistical binding.*

- **r -Robust Extractability** *There exists a QPT algorithm $\mathcal{SE}_r$ such that for any non-uniform QPT committer $\{C_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$, and r -round interactive machine B ,*

$$\{\text{OUT}_{\mathcal{SE}_r}(B(1^\lambda, z), \mathcal{SE}_r(1^\lambda, C_\lambda^*, \rho_\lambda))\}_{\lambda \in \mathbb{N}, z \in \{0,1\}^*} \stackrel{c}{\approx} \{\text{OUT}_{C_\lambda^*}(B(1^\lambda, z), C_\lambda^{*O^\infty}(\rho_\lambda))\}_{\lambda \in \mathbb{N}, z \in \{0,1\}^*}$$

(ε, r) -robustness is defined analogously when considering ε -indistinguishability instead of plain computational indistinguishability.

3.8 Rewinding-Safe Binding

We need an additional property for extractable commitments which we call *rewinding-safe binding*. This is analogous to the first-message binding property defined in [BLS22], but instead of asserting that the first message of the sender determines the committed value, we require that the first sender message that binds the value of the commitment occurs before extraction gadget of ExtCom . And we call such message the *binding* message. We give the definition as follow

Definition 21 (Rewinding-Safe Binding). *Let ExtCom be a extractable commitment scheme. We say ExtCom is rewinding-safe binding if it satisfies the following:*

1. *There exists a binding message, which is a single message sent by the sender that binds the value of the commitment. If there are multiple messages that binds the committed value, bind message is defined to be the first one.*
2. *Binding message occurs before the extraction gadget of `ExtCom`, namely during extraction, binding message is not rewound.*

4 Almost-Total Puzzle

4.1 Definition

We define a new notion called (post-quantum) almost-total puzzle, which will be used as a preamble in our construction of ppZKAoK in [Prot. 2](#) and public-coin non-malleable commitment in [Prot. 5](#) and [Prot. 6](#).

Definition 22 (Almost-Total Puzzle). *A (post-quantum) almost-total puzzle consists of a pair of PPT ITMs $\langle G, R \rangle$ and a deterministic polynomial-time algorithm `Verify` that satisfy the following syntax requirements.*

- **Puzzle Generation:** *Upon receiving the security parameter 1^λ as a common input, G and R interact with each other to generate a transcript τ (which encodes a puzzle). We denote this execution by $\tau \leftarrow \langle G, R \rangle(1^\lambda)$.*
- **Verification:** *`Verify` takes as input a transcript τ and a candidate solution s , and outputs a decision $d \in \{\top, \perp\}$*

The scheme satisfies the following requirements:

1. **Hard.** *For any non-uniform PPT/QPT puzzle receiver $\{R_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$, the following holds:*

$$\Pr \left[\text{Verify}(\tau, s) = \top : (\tau, s) \leftarrow \langle G, R_\lambda^*(\rho_\lambda) \rangle(1^\lambda) \right] \leq \text{negl}(\lambda) \quad (1)$$

where $(\tau, s) \leftarrow \langle G, R_\lambda^(\rho_\lambda) \rangle(1^\lambda)$ denotes the procedure where G and $R_\lambda^*(\rho_\lambda)$ interact to generate a transcript τ and R_λ^* outputs a string s .*

Intuitively, [Inequality \(1\)](#) means that no PPT/QPT malicious receiver R^ could find a puzzle solution s except for with negligible probability, even if R^* participates in the generation of the puzzle τ .*

2. **Almost-Total.** *For any unbounded-time puzzle sender G^* , the following holds:*

$$\Pr \left[\nexists s \text{ s.t. } \text{Verify}(\tau, s) = \top : \tau \leftarrow \langle G^*, R \rangle(1^\lambda) \right] \leq \text{negl}(\lambda). \quad (2)$$

3. **Public-Coin.** *During the puzzle generation procedure, all the messages sent by the honest puzzle sender G consist of public coins only.*

Note that [Def. 22](#) is post-quantum; a classical almost-total puzzle (ATP) can be defined in the same manner except that the **Hard** property (i.e., [Property 1](#)) only needs to hold against non-uniform PPT (instead of QPT) adversaries.

We remark that the existence of non-interactive post-quantum public-coin almost-total puzzles is equivalent to the existence of average-case quantum hardness in TFNP w.r.t. the uniform distribution.¹⁴ It is known that one-way permutations or public-coin collision-resistant hash functions

¹⁴ In the non-interactive setting, we can assume that the problem is “perfectly total”, i.e., the probability in [Inequality \(2\)](#) is 0, without loss of generality.

imply average-case quantum hardness in TFNP [Pap94], and the implication trivially extends to the post-quantum setting. Thus, we have the following lemma.

Lemma 2. *Assuming the existence of post-quantum public-coin collision-resistant hash functions or post-quantum one-way permutations, there exists a non-interactive post-quantum public-coin almost-total puzzle.*

Remark 5. Unruh [Unr16] pointed out that the “right” quantum analog of collision-resistance is what is called the *collapsing* property, which is stronger than post-quantum collision-resistance, in many settings. However, we stress that post-quantum collision-resistance, rather than the collapsing property, is sufficient in the above lemma as long as it is public-coin. We remark that post-quantum public-coin collision-resistant hash functions exist based on standard assumptions such as quantum hardness of the shortest integer solution (SIS) problem. On the other hand, there is no known candidate of post-quantum one-way permutations.

Unfortunately, it is unknown if OWFs imply average-case hardness in TFNP. Nonetheless, we show that OWFs imply two-round public-coin almost-total puzzles, using ideas inspired by [HNY17]. We formalize this result as the following Lem. 3, with a constructive proof in Sec. 4.2.

Lemma 3. *Assuming the existence of OWFs (resp. PQ-OWFs), there exists a two-round construction of public-coin almost-total puzzles (resp. PQ-ATPs).*

4.2 Construction

We construct a two-round public-coin almost-total puzzle from (PQ-)OWFs. Its description is given in Prot. 1. This construction is based on the following building block:

- A universal one-way hash function family $\mathcal{H} = \{H_\lambda\}_{\lambda \in \mathbb{N}}$ where

$$H_\lambda = \{h_k : \{0, 1\}^{\ell_{\text{in}}(\lambda)} \rightarrow \{0, 1\}^{\ell_{\text{out}}(\lambda)}\}_{k \in \{0, 1\}^{\ell_{\text{key}}(\lambda)}}.$$

Note that we have $\ell_{\text{out}}(\lambda) < \ell_{\text{in}}(\lambda)$ by the compressing property.

Protocol 1: Almost-Total Puzzle
Parameter: Let $n(\lambda) := \ell_{\text{key}}(\lambda) + \ell_{\text{in}}(\lambda) + \lambda$. Puzzle Generation: 1. The receiver R uniformly samples n random string $x'_1 \dots x'_n$ where $x'_i \xleftarrow{\\$} \{0, 1\}^{\ell_{\text{in}}(\lambda)}$ for each $i \in [n]$; R sends them to the generator G. 2. G samples a UOWHF key $k \xleftarrow{\\$} \{0, 1\}^{\ell_{\text{key}}(\lambda)}$ and a random string $x \xleftarrow{\\$} \{0, 1\}^{\ell_{\text{in}}(\lambda)}$ and sends them to R. Verification Algorithm $\text{Verify}(\tau, s)$: 1. Check the transcript τ: – if R's message is ill-formed (i.e., it is not of the form of $(x'_1, \dots, x'_n) \in (\{0, 1\}^{\ell_{\text{in}}(\lambda)})^{n(\lambda)}$), then output $\perp$; – if G's message is ill-formed (i.e., it is not of the form of $(k, x) \in \{0, 1\}^{\ell_{\text{key}}(\lambda)} \times \{0, 1\}^{\ell_{\text{in}}(\lambda)}$), then output $\top$.

Otherwise, parse $\tau = ((x'_1, \dots, x'_n), (k, x))$. 2. For each $i \in [n]$, compute $x_i = x'_i \oplus x$. 3. If there exists an $i \in [n]$ such that $(h_k(x_i) = h_k(s)) \wedge (x_i \neq s)$, then output $\top$. Otherwise, output $\perp$.
--

4.3 Proof of Properties

The public-coin property is straightforward from the description of [Prot. 1](#). So, we prove the *hard* property of our protocol first, which is established by the following lemma.

Lemma 4. *The protocol described in [Prot. 1](#) is hard (as per [Property 1](#) of [Def. 22](#)).*

Proof. We prove [Lem. 4](#) by contradiction. In the following, we assume for contradiction that there exists a non-uniform QPT/PPT puzzle receiver $R^* = \{R_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$ and a function $\delta(\lambda) = 1/\text{poly}(\lambda)$ such that for infinitely many $\lambda \in \mathbb{N}$,

$$\Pr \left[\text{Verify}(\tau, s) = 1 : (\tau, s) \leftarrow \langle G, R_\lambda^*(\rho_\lambda) \rangle(1^\lambda) \right] \geq \delta(\lambda). \quad (3)$$

Levering this $R_\lambda^*(\rho_\lambda)$, we construct in [Algo. 4.1](#) a (non-uniform) QPT/PPT algorithm $\mathcal{A} = \{\mathcal{A}_\lambda, \rho_\lambda\}_{\lambda \in \mathbb{N}}$ breaking the universal one-wayness of the UOWHF $\mathcal{H}$ (as per [Def. 3](#)):

Algorithm 4.1: $\mathcal{A}$ against UOWHF $\mathcal{A}$ internally emulates the algorithm $R_\lambda^*(\rho_\lambda)$ (dubbed R^* henceforth), while externally interacting with a challenger of the universal one-wayness of $\mathcal{H}$. In particular, $\mathcal{A}$ proceeds as follows. 1. $\mathcal{A}$ samples a string $x^* \xleftarrow{\\$} \{0, 1\}^{\ell_{\text{in}}(\lambda)}$ and sends it to the external challenger. 2. $\mathcal{A}$ receives a hash key k from the challenger. 3. $\mathcal{A}$ invokes R^* internally to receive n strings $x'_1 \dots x'_n$ from R^*. $\mathcal{A}$ then samples $i \xleftarrow{\\$} [n]$, sets $x := x^* \oplus x'_i$, and sends (k, x) to R^*. 4. $\mathcal{A}$ receives s from R^* and forward it to the challenger.
--

Clearly, $\mathcal{A}$ runs in polynomial time, and notice that the hash key k is sampled uniformly by the external challenger, and x^* is sampled uniformly which implies that x is also uniformly random. Therefore, from the R^* 's perspective, this is exactly the same execution as [Prot. 1](#), and thus it will output a valid solution s with probability at least $\delta(\lambda)$ due to our assumption in [Inequality \(3\)](#).

If s is a valid solution, then with probability $1/n$, $\mathcal{A}$ finds a collision for x^* (since we choose i randomly). Namely, $\mathcal{A}$ wins the universal one-wayness game with probability $\delta(\lambda)/n$. Since n is polynomial on λ , this leads to a contradiction against the security of the UOWHF $\mathcal{H}$. $\square$

Next, we prove the *almost-total* property of our protocol, which is formalized as the following [Lem. 5](#).

Lemma 5. *The protocol described in [Prot. 1](#) is almost-total (as per [Property 2](#) of [Def. 22](#)).*

Proof. For any $k \in \{0, 1\}^{\ell_{\text{key}}(\lambda)}$, since h_k is compressing, we have:

$$\Pr_{x_i \leftarrow \{0, 1\}^{\ell_{\text{in}}(\lambda)}} [\exists s \neq x_i \text{ s.t. } h_k(s) = h_k(x_i)] \geq \frac{1}{2}. \quad (4)$$

Thus, for any fixed $k \in \{0, 1\}^{\ell_{\text{key}}(\lambda)}$ and $x \in \{0, 1\}^{\ell_{\text{in}}(\lambda)}$, since $x_i = x \oplus x'_i$ is uniformly random when x'_i is uniformly random, we have:

$$\Pr_{x'_i \leftarrow \{0,1\}^{\ell_{\text{in}}(\lambda)}} [\exists s \neq x_i \text{ s.t. } h_k(s) = h_k(x_i) : x_i \leftarrow x \oplus x'_i] \geq \frac{1}{2}. \quad (5)$$

Negating [Inequality \(5\)](#), we get

$$\Pr_{x'_i \leftarrow \{0,1\}^{\ell_{\text{in}}(\lambda)}} [\forall s \neq x_i, h_k(s) \neq h_k(x_i) : x_i \leftarrow x \oplus x'_i] \leq \frac{1}{2}. \quad (6)$$

If we sample independently random x'_i for each $i \in [n]$, then the event in the above probability is independent for each $i \in [n]$. Thus, we have

$$\Pr_{(x'_1 \dots x'_n) \leftarrow \{0,1\}^{\ell_{\text{in}}(\lambda) \cdot n}} [\forall i \in [n], \forall s \neq x_i, h_k(s) \neq h_k(x_i)] \leq \left(\frac{1}{2}\right)^n \quad (7)$$

where $x_i := x \oplus x'_i$. By taking a union bound over all possible $k \in \{0, 1\}^{\ell_{\text{key}}(\lambda)}$ and $x \in \{0, 1\}^{\ell_{\text{in}}(\lambda)}$, we have

$$\begin{aligned} & \Pr_{x'_1 \dots x'_n \leftarrow \{0,1\}^{\ell_{\text{in}}(\lambda) \cdot n}} \left[\exists (k, x) \in \{0, 1\}^{\ell_{\text{key}}(\lambda)} \times \{0, 1\}^{\ell_{\text{in}}(\lambda)} \text{ s.t. } \forall i \in [n], \forall s \neq x_i, h_k(s) \neq h_k(x_i) \right] \\ & \leq \frac{2^{\ell_{\text{key}}(\lambda) + \ell_{\text{in}}(\lambda)}}{2^n} = 2^{-\lambda} \end{aligned} \quad (8)$$

where $x_i := x \oplus x'_i$ and the last equality follows from our parameter setting $n(\lambda) = \ell_{\text{key}}(\lambda) + \ell_{\text{in}}(\lambda) + \lambda$. Negating [Inequality \(8\)](#), we have

$$\begin{aligned} & \Pr_{x'_1 \dots x'_n \leftarrow \{0,1\}^{\ell_{\text{in}}(\lambda) \cdot n}} \left[\forall (k, x) \in \{0, 1\}^{\ell_{\text{key}}(\lambda)} \times \{0, 1\}^{\ell_{\text{in}}(\lambda)}, \exists i \in [n], \exists s \neq x_i \text{ s.t. } h_k(s) = h_k(x_i) \right] \\ & \geq 1 - 2^{-\lambda}. \end{aligned} \quad (9)$$

In words, for at least $(1 - 2^{-\lambda})$ -fraction of $(x'_1, \dots, x'_n)$ sampled by R , no matter how G^* chooses (k, x) , there is a solution s such that $\text{Verify}(\tau, s) = \top$ where $\tau = ((x'_1, \dots, x'_n), (k, x))$. This means that the almost-total property is satisfied. $\square$

The above completes the proof of [Lem. 3](#).

5 Post-Quantum ZKAoK with Preprocessing

In this section, we introduce the concept of interactive arguments with *preprocessing*. Essentially, these protocols are divided into two stages: a *preprocessing stage* and a *prove stage*. During the preprocessing stage, the prover and verifier interact without knowledge of the statement that needs to be proven. Once the statement is available, they engage in the prove stage to establish its truthfulness. This approach can be viewed as an extension of delayed-input protocols, where the instance is only required in the *final* round.

We first formalize the notion of zero-knowledge within this preprocessing framework, which we call *preprocessing zero-knowledge* (ppZK). We then demonstrate that ppZK leads to useful properties such as witness hiding and strong witness indistinguishability, all within the same preprocessing model.

5.1 Definition

We start by formally defining interactive arguments with preprocessing.

Syntax. An interactive argument with preprocessing consists of PPT ITMs P referred to as a prover and V referred to as a verifier. The interaction between P and V is divided into two stages, the preprocessing stage and the prove stage. In the preprocessing stage, P and V take the security parameter 1^λ as input, interact with each other, and generate a transcript $\mathbf{pp}$. We write $\mathbf{pp} \leftarrow \langle P, V \rangle_{\text{pre}}(1^\lambda)$ to mean this. In the prove stage, P and V take an instance x as well as $\mathbf{pp}$ as common inputs, P takes a witness w of x as an additional private input, and P and V interact with each other. At the end of the prove stage, V outputs $\top$ indicating acceptance or $\perp$ indicating rejection. We write $b \leftarrow \langle P(w), V \rangle_{\text{prove}}(\mathbf{pp}, x)$ to mean this where $b \in \{\top, \perp\}$ is V 's decision.

When we consider a malicious prover or verifier, we allow it to keep a state information from the preprocessing stage to the prove stage. Moreover, a malicious prover may adaptively choose the instance and output an arbitrary quantum state at the end of the prove stage. Specifically, for a malicious prover P^* with initial state ρ , we write $(\mathbf{pp}, x, \rho') \leftarrow \langle P^*(\rho), V \rangle_{\text{pre}}(1^\lambda)$ to mean the execution of the preprocessing stage between $P^*(\rho)$ and V on the security parameter 1^λ where $\mathbf{pp}$ is the transcript of the preprocessing stage, x is the statement chosen by P^* , and ρ' as the internal state of P^* at the end of the preprocessing stage, and $(\text{ST}'_{P^*}, b') \leftarrow \langle P^*(\rho'), V \rangle_{\text{prove}}(\mathbf{pp}, x)$ to mean the execution of the prove stage between $P^*(\rho')$ and V on the common inputs $\mathbf{pp}$ and x where ST_{P^*} is the final state of P^* and b is the decision output by V . Similarly, for a malicious verifier V^* with initial state ρ , we write $(\mathbf{pp}, \rho') \leftarrow \langle P, V^*(\rho) \rangle_{\text{pre}}(1^\lambda)$ to mean the execution of the preprocessing stage between P and $V^*(\rho)$ on the security parameter 1^λ where $\mathbf{pp}$ is the transcript of the preprocessing stage and ρ'_λ is the internal state of V^* at the end of the preprocessing stage, and $\langle P(w), V^*(\rho') \rangle_{\text{prove}}(\mathbf{pp}, x)$ to mean the execution of the prove stage between $P(w)$ and $V^*(\rho')$ on common inputs $\mathbf{pp}$ and x .

Definition 23 (PQ-ppZKAoK with EQPT_c-Simulation). A post-quantum preprocessing zero-knowledge arguments of knowledge (PQ-ppZKAoK) for an NP language $\mathcal{L}$ is an interactive argument $\langle P, V \rangle$ with preprocessing satisfying the following:

- **Completeness:** There exists a negligible function negl such that for any $\lambda \in \mathbb{N}$, $x \in L \cap \{0, 1\}^\lambda$, and $w \in \mathcal{R}_{\mathcal{L}}(x)$, it holds that

$$\Pr \left[\langle P(w), V \rangle_{\text{prove}}(\mathbf{pp}, x) = \top : \mathbf{pp} \leftarrow \langle P, V \rangle_{\text{pre}}(1^\lambda) \right] = 1 - \text{negl}(\lambda).$$

- **Preprocessing Zero-Knowledge:** There is an unbounded-time machine $\mathcal{S}_0$ with polynomial-size outputs and a QPT machine $\mathcal{S}_1$ such that for any non-uniform QPT malicious verifier $V^* = \{V_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$ and a family $\{(x_{\mathbf{pp}}, w_{\mathbf{pp}})\}_{\mathbf{pp}}$ such that $x_{\mathbf{pp}} \in L \cap \{0, 1\}^\lambda$ and $w_{\mathbf{pp}} \in R_L(x_{\mathbf{pp}})$,¹⁵ it holds that

$$\begin{aligned} & \{ \text{OUT}_{V_\lambda^*} \langle P(w_{\mathbf{pp}}), V_\lambda^*(\rho'_\lambda) \rangle_{\text{prove}}(\mathbf{pp}, x_{\mathbf{pp}}) : (\mathbf{pp}, \rho'_\lambda) \leftarrow \langle P, V_\lambda^*(\rho_\lambda) \rangle_{\text{pre}}(1^\lambda) \}_{\lambda \in \mathbb{N}} \\ & \stackrel{c}{\approx} \{ \mathcal{S}_1^{V_\lambda^*(\rho'_\lambda)}(\mathcal{S}_0(\mathbf{pp}), x_{\mathbf{pp}}) : (\mathbf{pp}, \rho'_\lambda) \leftarrow \langle P, V_\lambda^*(\rho_\lambda) \rangle_{\text{pre}}(1^\lambda) \}_{\lambda \in \mathbb{N}}. \end{aligned}$$

- **Adaptive AoK with EQPT_c-Simulation** There exists an EQPT_c oracle machine $\mathcal{WE}$, called the witness-extended emulator, such that for any non-uniform QPT machine $\{P_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$ the following conditions hold:

¹⁵ The subscript $\mathbf{pp}$ ranges all values that may be generated by the execution of $\langle P, V_\lambda^*(\rho_\lambda) \rangle_{\text{pre}}(1^\lambda)$.

1.

$$\left\{ (\text{ST}'_{\mathcal{P}^*}, b') : \begin{array}{l} (\text{pp}, x, \rho'_\lambda) \leftarrow \langle P_\lambda^*(\rho_\lambda), V \rangle_{\text{pre}}(1^\lambda); \\ (\text{ST}'_{\mathcal{P}^*}, b') \leftarrow \langle P_\lambda^*(\rho'_\lambda), V \rangle_{\text{prove}}(\text{pp}, x) \end{array} \right\}_{\lambda \in \mathbb{N}}$$

$$\stackrel{c}{\approx} \left\{ (\text{ST}'_{\mathcal{P}^*}, b') : \begin{array}{l} (\text{pp}, x, \rho'_\lambda) \leftarrow \langle P_\lambda^*(\rho_\lambda), V \rangle_{\text{pre}}(1^\lambda); \\ (\text{ST}'_{\mathcal{P}^*}, b', w') \leftarrow \mathcal{WE}^{P_\lambda^*(\rho'_\lambda)}(x) \end{array} \right\}_{\lambda \in \mathbb{N}},$$

$$2. \Pr \left[b' = \top \wedge w' \notin \mathcal{R}_{\mathcal{L}}(x) : \begin{array}{l} (\text{pp}, x, \rho'_\lambda) \leftarrow \langle P_\lambda^*(\rho_\lambda), V \rangle_{\text{pre}}(1^\lambda); \\ (\text{ST}'_{\mathcal{P}^*}, b', w') \leftarrow \mathcal{WE}^{P_\lambda^*(\rho'_\lambda)}(x) \end{array} \right] = 0.$$

Remark 6. In the above definition, EQPT_c-simulation only appears in the AoK property, and the zero-knowledge property is defined by a pair of unbounded-time and (strict) QPT simulators.

5.2 Implication

The processing zero-knowledge property defined in Def. 23 is sufficient in many settings where an instance is given to the verifier after preprocessing. For example, it implies strong witness indistinguishability and witness hiding in the preprocessing setting as defined below:

Definition 24 (Preprocessing Strong Witness Indistinguishability). *An interactive argument $\langle P, V \rangle$ with preprocessing for an NP language $\mathcal{L}$ satisfies preprocessing strong witness indistinguishability if it satisfies the following:*

- **Preprocessing Strong Witness Indistinguishability:** *For any pair of families of distributions $\{(\mathcal{X}_{0,\lambda,\text{pp}}, \mathcal{W}_{0,\lambda,\text{pp}})\}_{\lambda,\text{pp}}$ and $\{(\mathcal{X}_{1,\lambda,\text{pp}}, \mathcal{W}_{1,\lambda,\text{pp}})\}_{\lambda,\text{pp}}$ where $(\mathcal{X}_{b,\lambda,\text{pp}}, \mathcal{W}_{b,\lambda,\text{pp}})$ is a joint distribution supported on $\mathcal{R}_{\mathcal{L}} \cap (\{0,1\}^\lambda \times \{0,1\}^*)$ and $\{\mathcal{X}_{0,\lambda,\text{pp}}\}_{\lambda,\text{pp}} \stackrel{c}{\approx} \{\mathcal{X}_{1,\lambda,\text{pp}}\}_{\lambda,\text{pp}}$, and for any non-uniform QPT machine $\{V_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$, it holds that*

$$\left\{ \text{OUT}_{V_\lambda^*} \langle P(w_0), V_\lambda^*(\rho'_\lambda) \rangle_{\text{prove}}(\text{pp}, x_0) : \begin{array}{l} (\text{pp}, \rho'_\lambda) \leftarrow \langle P, V_\lambda^*(\rho_\lambda) \rangle_{\text{pre}}(1^\lambda) \\ (x_0, w_0) \leftarrow (\mathcal{X}_{0,\lambda,\text{pp}}, \mathcal{W}_{0,\lambda,\text{pp}}) \end{array} \right\}_{\lambda \in \mathbb{N}}$$

$$\stackrel{c}{\approx} \left\{ \text{OUT}_{V_\lambda^*} \langle P(w_1), V_\lambda^*(\rho'_\lambda) \rangle_{\text{prove}}(\text{pp}, x_1) : \begin{array}{l} (\text{pp}, \rho'_\lambda) \leftarrow \langle P, V_\lambda^*(\rho_\lambda) \rangle_{\text{pre}}(1^\lambda) \\ (x_1, w_1) \leftarrow (\mathcal{X}_{1,\lambda,\text{pp}}, \mathcal{W}_{1,\lambda,\text{pp}}) \end{array} \right\}_{\lambda \in \mathbb{N}}.$$

Definition 25 (Preprocessing Witness Hiding). *An interactive argument $\langle P, V \rangle$ with preprocessing for an NP language $\mathcal{L}$ satisfies preprocessing witness hiding if it satisfies the following:*

- **Preprocessing Witness Hiding:** *For any non-uniform QPT machine $\{V_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$, there is a non-uniform QPT machine $\{M_\lambda, \sigma_\lambda\}_{\lambda \in \mathbb{N}}$ such that for any family of distributions $\{(\mathcal{X}_\lambda, \mathcal{W}_\lambda)\}_\lambda$ where $(\mathcal{X}_\lambda, \mathcal{W}_\lambda)$ is a joint distribution supported on $\mathcal{R}_{\mathcal{L}} \cap (\{0,1\}^\lambda \times \{0,1\}^*)$, it holds that*

$$\Pr \left[\text{OUT}_{V_\lambda^*} \langle P(w), V_\lambda^*(\rho'_\lambda) \rangle_{\text{prove}}(\text{pp}, x) \in \mathcal{R}_{\mathcal{L}}(x) : \begin{array}{l} (\text{pp}, \rho'_\lambda) \leftarrow \langle P, V_\lambda^*(\rho_\lambda) \rangle_{\text{pre}}(1^\lambda) \\ (x, w) \leftarrow (\mathcal{X}_\lambda, \mathcal{W}_\lambda) \end{array} \right]$$

$$\leq \Pr[M_\lambda(\sigma_\lambda, x) \in \mathcal{R}_{\mathcal{L}}(x) : (x, w) \leftarrow (\mathcal{X}_\lambda, \mathcal{W}_\lambda)] + \text{negl}(\lambda).$$

Comparison with definitions without preprocessing. One might naturally assume sWI/WH with preprocessing are stronger notions that imply their standard counterparts, as the input can be delayed. However, in terms of existence, it turns out that standard sWI/WH are actually **stronger** than their preprocessing counterparts, contrary to intuition. If we have a standard sWI/WH protocol, we can treat all rounds as part of the prove stage, making the preprocessing stage effectively

null. This would naturally imply the corresponding preprocessing sWI/WH properties. However, the converse does not hold. Consider a preprocessing sWI/WH argument: if we treat it as a standard interactive argument by combining the preprocessing and prove stages into a single stage, it does not necessarily satisfy standard sWI/WH. Intuitively, preprocessing sWI/WH only guarantees security as long as the messages in the preprocessing stage do not depend on the instance to be proven. This condition restricts the verifier's behavior, and thus the security no longer holds if the verifier's preprocessing messages are allowed to depend on the instance.

Although standard sWI/WH and preprocessing sWI/WH cannot be directly compared, we observe that a preprocessing interactive argument can be viewed as a standard interactive argument without preprocessing (as defined in [Sec. 3.4](#)) by merging the preprocessing and prove stages into a single protocol. The AoK property in the preprocessing setting (as per [Def. 23](#)) implies that in the without-preprocessing setting (as per [Def. 9](#)). This is because the preprocessing AoK property is stronger, as it captures a cheating prover who adaptively selects the statement after completing the preprocessing stage. On the other hand, the preprocessing zero-knowledge property (as per [Def. 23](#)) does not imply the standard zero-knowledge property in the without-preprocessing setting, since the preprocessing part of the simulator $\mathcal{S}_0$ in our definition needs unbounded time. Nonetheless, we remark that the preprocessing zero-knowledge property (as per [Def. 23](#)) does imply witness indistinguishability in the without-preprocessing setting (as per [Def. 5](#)). This follows from the well-known fact that witness indistinguishability is equivalent to zero-knowledge with unbounded-time simulation. Based on the above observation, we have the following lemma.

Lemma 6. *For a PQ-ppZKAoK with EQPT_c -simulation, if we regard the merge of preprocessing and prove stages as a single protocol without preprocessing, then the resulting protocol is a PQ-WIAoK with EQPT_c -simulation.*

It is straightforward to see the following lemma.

Lemma 7. *Preprocessing zero-knowledge ([Def. 23](#)) implies preprocessing strong witness indistinguishability ([Def. 24](#)) and preprocessing witness hiding ([Def. 25](#)).*

Proof. Let $(\mathcal{S}_0, \mathcal{S}_1)$ be the simulator for the preprocessing zero-knowledge property. Let $\{V_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$ be a non-uniform QPT malicious verifier. We prove preprocessing strong witness indistinguishability and preprocessing witness hiding below.

Preprocessing Strong Witness Indistinguishability. By the preprocessing zero-knowledge property, we have the following for $b \in \{0, 1\}$:

$$\begin{aligned} & \left\{ \text{OUT}_{V_\lambda^*} \langle P(w_b), V_\lambda^*(\rho'_\lambda) \rangle_{\text{prove}}(\text{pp}, x_b) : \begin{array}{l} (\text{pp}, \rho'_\lambda) \leftarrow \langle P, V_\lambda^*(\rho_\lambda) \rangle_{\text{pre}}(1^\lambda) \\ (x_b, w_b) \leftarrow (\mathcal{X}_{b,\lambda,\text{pp}}, \mathcal{W}_{b,\lambda,\text{pp}}) \end{array} \right\}_{\lambda \in \mathbb{N}} \\ & \stackrel{c}{\approx} \left\{ \mathcal{S}_1^{V_\lambda^*(\rho'_\lambda)}(\mathcal{S}_0(\text{pp}), x_b) : \begin{array}{l} (\text{pp}, \rho'_\lambda) \leftarrow \langle P, V_\lambda^*(\rho_\lambda) \rangle_{\text{pre}}(1^\lambda) \\ (x_b, w_b) \leftarrow (\mathcal{X}_{b,\lambda,\text{pp}}, \mathcal{W}_{b,\lambda,\text{pp}}) \end{array} \right\}_{\lambda \in \mathbb{N}}. \end{aligned}$$

By the assumption that $\{\mathcal{X}_{0,\lambda,\text{pp}}\}_{\lambda,\text{pp}} \stackrel{c}{\approx} \{\mathcal{X}_{1,\lambda,\text{pp}}\}_{\lambda,\text{pp}}$, we have

$$\begin{aligned} & \left\{ \mathcal{S}_1^{V_\lambda^*(\rho'_\lambda)}(\mathcal{S}_0(\text{pp}), x_0) : \begin{array}{l} (\text{pp}, \rho'_\lambda) \leftarrow \langle P, V_\lambda^*(\rho_\lambda) \rangle_{\text{pre}}(1^\lambda) \\ (x_0, w_0) \leftarrow (\mathcal{X}_{0,\lambda,\text{pp}}, \mathcal{W}_{0,\lambda,\text{pp}}) \end{array} \right\}_{\lambda \in \mathbb{N}} \\ & \stackrel{c}{\approx} \left\{ \mathcal{S}_1^{V_\lambda^*(\rho'_\lambda)}(\mathcal{S}_0(\text{pp}), x_1) : \begin{array}{l} (\text{pp}, \rho'_\lambda) \leftarrow \langle P, V_\lambda^*(\rho_\lambda) \rangle_{\text{pre}}(1^\lambda) \\ (x_1, w_1) \leftarrow (\mathcal{X}_{1,\lambda,\text{pp}}, \mathcal{W}_{1,\lambda,\text{pp}}) \end{array} \right\}_{\lambda \in \mathbb{N}}. \end{aligned}$$

We remark that “ $\overset{c}{\approx}$ ” means computational indistinguishability against *non-uniform* QPT distinguishers, and thus we can reduce the above to $\{\mathcal{X}_{0,\lambda,\text{pp}}\}_{\lambda,\text{pp}} \overset{c}{\approx} \{\mathcal{X}_{1,\lambda,\text{pp}}\}_{\lambda,\text{pp}}$ by regarding $\mathcal{S}_0(\text{pp})$ as a non-uniform advice. By combining the above, preprocessing strong witness indistinguishability follows.

Preprocessing Witness Hiding. By the preprocessing zero-knowledge property, it holds that

$$\left| \Pr \left[\text{OUT}_{V_\lambda^*} \langle P(w), V_\lambda^*(\rho'_\lambda) \rangle_{\text{prove}}(\text{pp}, x) \in \mathcal{R}_\mathcal{L}(x) : \begin{array}{l} (\text{pp}, \rho'_\lambda) \leftarrow \langle P, V_\lambda^*(\rho_\lambda) \rangle_{\text{pre}}(1^\lambda) \\ (x, w) \leftarrow (\mathcal{X}_\lambda, \mathcal{W}_\lambda) \end{array} \right] \right. \\ \left. - \Pr \left[\mathcal{S}_1^{V_\lambda^*(\rho'_\lambda)}(\mathcal{S}_0(\text{pp}), x) \in \mathcal{R}_\mathcal{L}(x) : \begin{array}{l} (\text{pp}, \rho'_\lambda) \leftarrow \langle P, V_\lambda^*(\rho_\lambda) \rangle_{\text{pre}}(1^\lambda) \\ (x, w) \leftarrow (\mathcal{X}_\lambda, \mathcal{W}_\lambda) \end{array} \right] \right| \leq \text{negl}(\lambda).$$

We define a non-uniform QPT machine $\{M_\lambda, \sigma_\lambda\}_{\lambda \in \mathbb{N}}$ as follows:

- $\sigma_\lambda := (\mathcal{S}_0(\text{pp}), \rho'_\lambda)$ where $(\text{pp}, \rho'_\lambda) \leftarrow \langle P, V_\lambda^*(\rho_\lambda) \rangle_{\text{pre}}(1^\lambda)$;
- M_λ takes $\sigma_\lambda = (\mathcal{S}_0(\text{pp}), \rho'_\lambda)$ as advice and x as input, runs $\mathcal{S}_1^{V_\lambda^*(\rho'_\lambda)}(\mathcal{S}_0(\text{pp}), x)$, and outputs its output.

Clearly, the second probability in the above inequality is equal to

$$\Pr[M_\lambda(\sigma_\lambda, x) \in \mathcal{R}_\mathcal{L}(x) : (x, w) \leftarrow (\mathcal{X}_\lambda, \mathcal{W}_\lambda)].$$

Thus, the preprocessing witness hiding follows. □

5.3 Public-Coin, Five-Round Construction

We now present a five-round, public-coin construction of PQ-ppZKAoK with EQPT_c -simulation, assuming the existence of post-quantum *one-way functions*. Our protocol makes use of the following building blocks:

- The two-round public-coin almost-total puzzle constructed in [Prot. 1](#); We write **Verify** to mean its verification algorithm.
- Naor’s commitment **Com** (more accurately, its post-quantum version instantiated from PQ-OWFs); We use β to denote the first message for Naor’s commitment, and $\text{Com}_\beta(m; r)$ to denote the second message where a string m is committed using randomness r .
- A four-round public-coin witness-indistinguishable proof of knowledge **WIPoK** with EQPT_m -Guaranteed-Extraction ([Def. 11](#)).

Protocol 2: Post-Quantum Constant Round Public-Coin ppZKAoK with EQPT_c -simulation
Let $\mathcal{L}$ be an NP language, $x \in \mathcal{L} \cap \{0, 1\}^\lambda$ and $w \in \mathcal{R}_\mathcal{L}(x)$.
Preprocessing Stage:
1. Prover P (acting as the puzzle receiver R_{PUZ}) sends to the verifier V (acting as the puzzle sender G_{PUZ}) the first message of the almost-total puzzle protocol.
2. V sends to P the second message of the almost-total puzzle protocol, the first message β for Naor’s commitment, and the first message v of WIPoK . Let τ be the transcript of the almost-total puzzle protocol.
At this point, P and V share the transcript $\text{pp} = (\beta, v, \tau)$ of the preprocessing stage.

Prove Stage:

1. **(Commit to Witness.)** P samples a random tape r and sends $c = \text{Com}_\beta(w; r)$ to V ;
2. **(WiPoK.)** P and V run the rest of execution of WiPoK where P proves to V that either he “knows” a solution for τ or c is a commitment of a valid witness $w \in \mathcal{R}_\mathcal{L}(x)$. Formally, P proves that $(\tau, c) \in \mathcal{L}_{\text{puz}} \vee \mathcal{L}_\beta^x$, and $\mathcal{L}_{\text{puz}} \vee \mathcal{L}_\beta^x$ denotes the OR-composed language, where

$$\mathcal{L}_{\text{puz}} := \{\tau \mid \exists s \text{ s.t. } \text{Verify}(\tau, s) = 1\}. \quad (10)$$

and

$$\mathcal{L}_\beta^x := \{c \mid \exists (w, r) \text{ s.t. } c = \text{Com}_\beta(w; r) \wedge w \in \mathcal{R}_\mathcal{L}(x)\}. \quad (11)$$

Note that P uses the (w, r) as the witness when executing this WiPoK.

Theorem 4. *Prot. 2 is a public-coin PQ-ppZKAoK with EQPT_c -simulation (as per Def. 23). Its preprocessing stage is two-round and prove stage is three-round.*

Completeness, the public-coin property, and the round complexity are straightforward from the description of Prot. 2 noting that the commitment in Step 1 of the prove stage can be sent in parallel to the first message in Step 2 of the prove stage (which is the second-round message of WiPoK). The rest of the proof of Thm. 4 (preprocessing zero-knowledge and adaptive AoK with EQPT_c -simulation) is given in Sec. 5.4 and 5.5.

By Thm. 4 and Lem. 6, we obtain the following corollary.

Corollary 6. *Assuming the existence of PQ-OWFs, there is a five-round public-coin PQ-WIAoK with EQPT_c -simulation (as per Def. 9).*

Reducing the round complexity from stronger assumptions. We remark that the round complexity of the preprocessing stage can be reduced by one if we use non-interactive almost-total puzzles instead of the two-round one. By Lem. 2, we have the following theorem:

Theorem 5. *Assuming the existence of post-quantum collision-resistant hash functions or post-quantum one-way permutations, there is a public-coin PQ-ppZKAoK with EQPT_c -simulation (as per Def. 23) of which preprocessing stage is one-round and prove stage is three-round.*

Thm. 5 and Lem. 6 imply the following corollary.

Corollary 7. *Assuming the existence of post-quantum public-coin collision-resistant hash functions or post-quantum one-way permutations, there is a four-round public-coin PQ-WIAoK with EQPT_c -simulation (as per Def. 9).*

5.4 Proof of Preprocessing Zero-Knowledge

Lemma 8. *The protocol described in Prot. 2 satisfies the preprocessing zero-knowledge property.*

Proof. Let $V^* = \{V_\lambda^*, \rho_\lambda\}_\lambda$ be a non-uniform QPT adversary. We consider the following sequence of hybrids.

- Hybrid $H_1^{V^*}(\lambda)$: This is the real experiment that simulates the interaction between $P(x, w)$ and V^* . That is, it runs $(\text{pp}, \rho'_\lambda) \leftarrow \langle P, V_\lambda^*(\rho_\lambda) \rangle_{\text{pre}}(1^\lambda)$ and $\langle P(w_{\text{pp}}), V_\lambda^*(\rho'_\lambda) \rangle_{\text{prove}}(\text{pp}, x_{\text{pp}})$ and outputs the final state of V^* .

- Hybrid $H_2^{V*}(\lambda)$: This is identical to $H_1^{V*}(\lambda)$ except that it finds the lexicographically smallest solution s for τ by *brute-force* where τ is the transcript of the almost-total puzzle contained in **pp**. If a solution does not exist, it outputs $\perp$, and otherwise works similarly to the previous hybrid. We note that this experiment obtains s but does not use it. By the almost-total property, the probability that this hybrid outputs $\perp$ is negligible. Thus, we have

$$\{H_1^{V*}(\lambda)\}_{\lambda \in \mathbb{N}} \stackrel{s}{\approx} \{H_2^{V*}(\lambda)\}_{\lambda \in \mathbb{N}}. \quad (12)$$

- Hybrid $H_3^{V*}(\lambda)$: This is identical to $H_2^{V*}(\lambda)$ except that it uses s obtained by brute-force as the witness in WIPoK. By the (non-uniform) WI property of WIPoK, we have

$$\{H_2^{V*}(\lambda)\}_{\lambda \in \mathbb{N}} \stackrel{c}{\approx} \{H_3^{V*}(\lambda)\}_{\lambda \in \mathbb{N}}. \quad (13)$$

We remark that the brute-force is done before receiving any prover message of WIPoK and thus we can still rely on the WI property against non-uniform adversaries by treating **pp**, s , and the verifier's internal state right after finishing the preprocessing stage as non-uniform advice.

- Hybrid $H_4^{V*}(\lambda)$: This is identical to $H_3^{V*}(\lambda)$ except that it commits to 0 instead of the witness in [Step 1](#). By the (non-uniform) hiding property of **Com**, we have

$$\{H_3^{V*}(\lambda)\}_{\lambda \in \mathbb{N}} \stackrel{c}{\approx} \{H_4^{V*}(\lambda)\}_{\lambda \in \mathbb{N}}. \quad (14)$$

One can see that $H_4^{V*}(\lambda)$ corresponds to the view simulated by the following simulator $\mathcal{S} = (\mathcal{S}_0, \mathcal{S}_1)$.

- $\mathcal{S}_0(\mathbf{pp})$: Parse $\mathbf{pp} = (\beta, v, \tau)$ and find the lexicographically smallest solution s of τ by brute-force. If a solution does not exist, set $s := \perp$. Output s .
- $\mathcal{S}_1^{V_\lambda^*(\rho'_\lambda)}(s, x_{\mathbf{pp}})$: If $s = \perp$, output $\perp$. Otherwise, interact with $V_\lambda^*(\rho'_\lambda)$ in the prove stage where it uses s as the witness for WIPoK.

The computational indistinguishability between $H_1^{V*}(\lambda)$ and $H_4^{V*}(\lambda)$ directly means the preprocessing zero-knowledge property. This completes the proof of [Lem. 8](#). $\square$

5.5 Proof of Adaptive AoK with EQPT_c-Simulation

Lemma 9. *The protocol described in [Prot. 2](#) satisfies adaptive AoK with EQPT_c-simulation.*

Proof. Let $\mathcal{GE}$ be the EQPT_m guaranteed extractor of WIPoK satisfying the property of [Def. 11](#). Let $U_{\mathcal{GE}}^{P_\lambda^*}(v, a, x)$ be a unitary dilation of $\mathcal{GE}^{P_\lambda^*}(v, a, x)$ that acts on the prover's internal register **P** and an ancilla register **Anc**. That is, it is defined in such a way that $\mathcal{GE}^{P_\lambda^*}(v, a, x)$ is equivalent to the following procedure:

1. Initialize **P** to be ρ_λ and **Anc** to be $|0\dots 0\rangle$.
2. Apply $U_{\mathcal{GE}}^{P_\lambda^*}(v, a, x)$.
3. Measure a designated subregister **W** of **Anc** to obtain w and discard **Anc**.

Moreover, we assume that **W** consists of subregisters $(\mathbf{W}_1, \mathbf{W}_2)$, and if the extracted witness is $(w, r) \in \mathcal{R}_{\mathcal{L}_\beta^x}(c)$, then w and r are written in $\mathbf{W}_1$ and $\mathbf{W}_2$, respectively. Then we construct the witness-extended emulator $\mathcal{WE}$ as follows.

$\mathcal{WE}^{P_\lambda^*}(x)$:

1. Run the preprocessing stage with P_λ^* while playing the role of the honest verifier. Let $\mathbf{pp} = (\beta, v, \tau)$ be the transcript of the preprocessing stage.
2. Receive $\mathbf{c}$ and the second-round message a of WIPoK from P_λ^* .
3. Generate a uniform superposition $\sum_{r \in R} |r\rangle$ over the possible values of the third-round message of WIPoK, and coherently run P_λ^* to generate the fourth-round message in superposition.
4. Measure the decision of the verifier of WIPoK on the transcript formed by v , a , and the values in the third-round and fourth-round message registers of WIPoK w.r.t. the instance $(\tau, \mathbf{c})$ (without measuring the values of the third-round and fourth messages themselves). Let b be the measurement outcome.
5. If $b = \perp$, then measure the transcript, let $\mathbf{ST}'_{P^*}$ be the internal state of P_λ^* at this point, and output $(\mathbf{ST}'_{P^*}, b = \perp, w = \perp)$ and halt.
6. If $b = \top$, do the following:
 - (a) Let $\mathbf{P}$ be the internal register of P_λ^* and initialize $\mathbf{Anc}$ to be $|0\dots 0\rangle$. Apply $U_{\mathcal{GE}}^{P_\lambda^*}(v, a, x)$.
 - (b) Measure the subregister $\mathbf{W}_1$ of $\mathbf{Anc}$ to obtain w .
 - (c) Apply $U_{\mathcal{GE}}^{P_\lambda^*}(v, a, x)^\dagger$.
 - (d) Measure the transcript, let $\mathbf{ST}'_{P^*}$ be the internal state of P_λ^* at this point, and output $(\mathbf{ST}'_{P^*}, b = \top, w)$.

Since $\mathcal{GE}$ is an EQPT_m machine, $\mathcal{WE}$ is an EQPT_c machine ([Thm. 2](#)).

We say that β is good if for any $\mathbf{c}$, there is at most one m such that $\mathbf{c} = \text{Com}_\beta(m; r)$ for some r . We denote such m by $\text{val}_\beta(\mathbf{c})$ where if there is no such m , then we define $\text{val}_\beta(\mathbf{c}) := \perp$. By the statistical binding property of Naor's commitment, β is good except for a negligible probability. We show the following claim.

Claim 6. *For any non-uniform QPT machine $\{P_\lambda^*, \rho_\lambda\}_\lambda$, instance $x \in \{0, 1\}^\lambda$, and good β , it holds that*

$$\Pr[b = \top \wedge (w \neq \text{val}_\beta(\mathbf{c}) \vee w \notin \mathcal{R}_{\mathcal{L}}(x))] \leq \text{negl}(\lambda)$$

where the probability is taken over the execution of $\mathcal{WE}^{P_\lambda^*(\rho_\lambda)}(x)$ with the fixed β .

Proof of [Thm. 6](#). Let $\widetilde{\mathcal{WE}}^{P_\lambda^*(\rho_\lambda)}(x)$ be the same as $\mathcal{WE}^{P_\lambda^*(\rho_\lambda)}(x)$ except that it measures the whole $\mathbf{W}$ (rather than only $\mathbf{W}_1$) in [Step 6b](#). Let $\tilde{w}$ be the measurement outcome. By the guaranteed extraction property, it holds that

$$\Pr[b = \top \wedge \tilde{w} \notin (\mathcal{R}_{\mathcal{L}_{\text{puz}}}(\tau) \cup \mathcal{R}_{\mathcal{L}_\beta^x}(\mathbf{c}))] \leq \text{negl}(\lambda) \quad (15)$$

where the probability is taken over the execution of $\widetilde{\mathcal{WE}}^{P_\lambda^*(\rho_\lambda)}(x)$. Moreover, by the hardness of the almost-total puzzle, it holds that

$$\Pr[b = \top \wedge \tilde{w} \in \mathcal{R}_{\mathcal{L}_{\text{puz}}}(\tau)] \leq \text{negl}(\lambda) \quad (16)$$

where the probability is taken over the execution of $\widetilde{\mathcal{WE}}^{P_\lambda^*(\rho_\lambda)}(x)$. Indeed, this can be seen as follows: Suppose that we have $\Pr[b = \top \wedge \tilde{w} \in \mathcal{R}_{\mathcal{L}_{\text{puz}}}(\tau)] \geq 1/p(\lambda)$ for some polynomial p and infinitely many λ . Since $\mathcal{GE}^{P_\lambda^*(\rho_\lambda)}(v, a, x)$ runs in EQPT_m , a straightforward reduction gives an EQPT_m

algorithm that solves the puzzle with the same probability.¹⁶ By an appropriate truncation, this gives a (strict) QPT algorithm that solves the with probability at least $1/(2p(\lambda))$ for infinitely many λ . This contradicts the hardness of the almost-total puzzle. Thus, Eq. (16) holds.

Combining Eq. (15) and (16), we have

$$\Pr[b = \top \wedge \tilde{w} \notin \mathcal{R}_{\mathcal{L}_\beta^x}(c)] \leq \text{negl}(\lambda) \quad (17)$$

where the probability is taken over the execution of $\widetilde{\mathcal{WE}}^{P_\lambda^*}(\rho_\lambda)(x)$. By the definitions of $\widetilde{\mathcal{WE}}$ and $\mathcal{L}_\beta^x$ and the assumption that β is good, Eq. (17) immediately implies Thm. 6. $\square$

The second requirement of the AoK with EQPT_c-simulation property (Condition 2 of Def. 9) immediately follows from Thm. 6 and the fact that an overwhelming fraction of β is good. To show the first requirement (Condition 1 of Def. 9), again by Thm. 6 and the fact that an overwhelming fraction of β is good, the measurement in Step 6b of the execution of $\mathcal{WE}^{P_\lambda^*}(\rho_\lambda)(x)$ results in the fixed value $\text{val}_\beta(c)$ except for a negligible probability. Thus, by the gentle measurement lemma, Steps 6a to 6c only negligibly changes the internal state of P_λ^* . Moreover if we remove these steps, then ST'_{P^*} is exactly identical to that in the real execution. This implies that the first requirement is also satisfied. This completes the proof of Lem. 9. $\square$

6 (Classical) Delayed-Input Non-Uniform ZKAoK

In this section, we introduce the concept of delayed-input non-uniform zero-knowledge (dinuZK), which we believe is significantly stronger than delayed-input strong WI. Delayed-input non-uniform ZK is defined similarly to the normal ZK except that the malicious verifier should be delayed-input (i.e., takes the instance at the last round) and the simulator can be a non-uniform machine that depends on the verifier.

6.1 Definition

Definition 26 (Delayed-input non-uniform ZK). *A delayed-input protocol $\langle P, V \rangle$ (Def. 6) satisfies delayed-input non-uniform ZK if for any non-uniform deterministic poly-time malicious delayed-input verifier V^* , there is a non-uniform PPT simulator S such that*

$$\{\text{View}_{V^*}(P(w), V^*)(x)\}_{\lambda, x, w} \stackrel{c}{\approx} \{S(x)\}_{\lambda, x, w}$$

where $\lambda \in \mathbb{N}$, $x \in L \cap \{0, 1\}^\lambda$, and $w \in R_L(x)$.

6.2 Implication

The delayed-input non-uniform zero-knowledge property defined in Def. 26 is sufficient in many settings where an instance is given to the verifier in the last round. For example, it implies delayed-input strong witness indistinguishability and delayed-input witness hiding (against non-uniform adversaries).

Lemma 10. *Delayed-input non-uniform zero-knowledge (Def. 26) implies delayed-input strong witness indistinguishability (Def. 7) and delayed-input witness hiding (Def. 8).*

¹⁶ Though $\widetilde{\mathcal{WE}}^{P_\lambda^*}(\rho_\lambda)(x)$ is an EQPT_c machine, we can generate b and $\tilde{w}$ from the same distribution in EQPT_m by running $\mathcal{GE}^{P_\lambda^*}(\rho_\lambda)(v, a, x)$ in Step 6a instead of applying its unitary dilation $U_{\mathcal{GE}}^{P_\lambda^*}(v, a, x)$. Note that we do not need to run the rest of the steps since b and $\tilde{w}$ are obtained at this point.

Proof. We prove delayed-input strong witness indistinguishability and delayed-input witness hiding below.

Delayed-Input Strong Witness Indistinguishability. Let V^* be a deterministic polynomial-time malicious verifier for the delayed-input strong WI property, which chooses the best randomness (randomness that maximize the winning probability) as its non-uniform advice. Let $\mathcal{S}$ be the non-uniform simulator for the delayed-input non-uniform zero-knowledge property, which chooses its non-uniform advice based on V^* . By the delayed-input non-uniform zero-knowledge property, we have the following for $b \in \{0, 1\}$:

$$\begin{aligned} & \{\text{View}_{V^*}(P(w), V^*)(x_b) : (x_b, w_b) \leftarrow (\mathcal{X}_{b,\lambda}, \mathcal{W}_{b,\lambda})\}_{\lambda \in \mathbb{N}} \\ & \stackrel{c}{\approx} \{\mathcal{S}(x_b) : (x_b, w_b) \leftarrow (\mathcal{X}_{b,\lambda}, \mathcal{W}_{b,\lambda})\}_{\lambda \in \mathbb{N}} \end{aligned}$$

By the assumption that $\{\mathcal{X}_{0,\lambda}\}_{\lambda \in \mathbb{N}} \stackrel{c}{\approx} \{\mathcal{X}_{1,\lambda}\}_{\lambda \in \mathbb{N}}$, we have

$$\begin{aligned} & \{\mathcal{S}(x_0) : (x_0, w_0) \leftarrow (\mathcal{X}_{0,\lambda}, \mathcal{W}_{0,\lambda})\}_{\lambda \in \mathbb{N}} \\ & \stackrel{c}{\approx} \{\mathcal{S}(x_1) : (x_1, w_1) \leftarrow (\mathcal{X}_{1,\lambda}, \mathcal{W}_{1,\lambda})\}_{\lambda \in \mathbb{N}} \end{aligned}$$

We remark that “ $\stackrel{c}{\approx}$ ” means computational indistinguishability against *non-uniform* PPT distinguishers, and thus we can reduce the above to $\{\mathcal{X}_{0,\lambda}\}_{\lambda \in \mathbb{N}} \stackrel{c}{\approx} \{\mathcal{X}_{1,\lambda}\}_{\lambda \in \mathbb{N}}$ by regarding $\mathcal{S}(x_b)$ as a non-uniform advice. By combining the above, preprocessing strong witness indistinguishability follows.

Delayed-Input Witness Hiding. Let V^* be a deterministic polynomial-time malicious verifier for the delayed-input WH property, which chooses the best randomness (randomness that maximize the winning probability) as its non-uniform advice. Let $\mathcal{S}$ be the non-uniform simulator for the delayed-input non-uniform zero-knowledge property, which chooses its non-uniform advice based on V^* . By the delayed-input non-uniform zero-knowledge property, it holds that

$$\begin{aligned} & |\Pr[\text{View}_{V^*}(P(w), V^*)(x) \in \mathcal{R}_{\mathcal{L}}(x) : (x, w) \leftarrow (\mathcal{X}_{\lambda}, \mathcal{W}_{\lambda})] \\ & - \Pr[\mathcal{S}(x) \in \mathcal{R}_{\mathcal{L}}(x) : (x, w) \leftarrow (\mathcal{X}_{\lambda}, \mathcal{W}_{\lambda})]| \leq \text{negl}(\lambda). \end{aligned}$$

We define a non-uniform PPT machine M , which takes the non-uniform advice of $\mathcal{S}$ as its own non-uniform advice, runs $\mathcal{S}$ internally, and outputs its output. Clearly, the second probability in the above inequality is equal to $\Pr[M(a_{\lambda}, x) \in \mathcal{R}_{\mathcal{L}}(x) : (x, w) \leftarrow (\mathcal{X}_{\lambda}, \mathcal{W}_{\lambda})]$. Thus, the delayed-input witness hiding follows. □

6.3 Five-Round, Public-Coin Construction

We present in [Prot. 3](#) a five-round, public-coin construction of dinuZK, assuming the existence of OWFs.

Protocol 3: Five-Round Public-Coin Delayed-Input Non-Uniform ZKAoK

By treating the preprocessing and proof stages as a unified protocol, [Prot. 2](#) also results in a classical dinuZKAoK construction. It is crucial to highlight that, to maintain the delayed-input property of dinuZK, a special type of WIPoK is required—one that ensures the instance is not necessary for the prover’s messages before the final round. This type of WIPoK, referred to as

delayed-input witness-indistinguishable *adaptive* proofs of knowledge in the literature, can be constructed from OWFs [LS91].

6.4 Security Proof

Completeness, the public-coin property, and the round complexity are straightforward from the description of [Prot. 3](#). The AoK property and the dinuZK property are established in [Lem. 11](#) and [12](#) respectively.

Lemma 11. *The protocol described in [Prot. 3](#) satisfies knowledge soundness.*

Proof. Proof (Sketch). By the knowledge soundness of WIPoK, there exists an oracle-aided PPT algorithm $\mathcal{K}$, such that

$$\begin{aligned} & \Pr \left[w' \in (\mathcal{R}_{\mathcal{L}_{\text{puz}}}(\tau) \cup \mathcal{R}_{\mathcal{L}_{\beta}^x}(\mathbf{c})) : w' \leftarrow \mathcal{K}^{P^*}(x) \right] \\ & \geq \frac{1}{\text{poly}(\lambda)} \cdot \Pr[\text{OUT}_V \langle P^*, V \rangle(x) = \top]^d - \text{negl}(\lambda). \end{aligned} \quad (18)$$

where the probability is taken over the execution of $\mathcal{K}^{P^*}(x)$. Moreover, by the hardness of the almost-total puzzle, it holds that

$$\Pr \left[w' \in \mathcal{R}_{\mathcal{L}_{\text{puz}}}(\tau) : w' \leftarrow \mathcal{K}^{P^*}(x) \right] \leq \text{negl}(\lambda). \quad (19)$$

Otherwise a straightforward reduction gives an PPT algorithm that solves the almost-total puzzle. Combining [Eq. \(18\)](#) and [\(19\)](#), we have

$$\Pr \left[w' \in \mathcal{R}_{\mathcal{L}_{\beta}^x}(\mathbf{c}) : w' \leftarrow \mathcal{K}^{P^*}(x) \right] \geq \frac{1}{\text{poly}(\lambda)} \cdot \Pr[\text{OUT}_V \langle P^*, V \rangle(x) = \top]^d - \text{negl}(\lambda). \quad (20)$$

By the definitions of $\mathcal{K}$ and $\mathcal{L}_{\beta}^x$, [Eq. \(20\)](#) immediately implies [Lem. 11](#). $\square$

Lemma 12. *The protocol described above satisfies the delayed-input non-uniform zero-knowledge property.*

Proof. Let V^* be a non-uniform deterministic poly-time adversary, and let $\mathcal{S}$ be the non-uniform simulator that depends on V^* . Since V^* is deterministic, $\mathcal{S}$ samples a random preimage x_0^* , and incorporates V^* 's choice of UOWHF hash key k and the preimage x (possibly based on x_0^*) for [Step 2](#) of the almost-total puzzle into its non-uniform advice. $\mathcal{S}$ then uses brute-force to find a random collision x_1^* for x_0^* under k as part of the non-uniform advice. Note that if no collision exists for x_0^* , repeat the above procedure until a collision is found, and by the compressing property of UOWFH, such collision always exists. We then consider the following sequence of hybrids.

- Hybrid $H_1^{V^*}(\lambda, x, w)$: This is the real experiment that simulates the interaction between $P(x, w)$ and V^* , and outputs $\text{View}_{V^*} \langle P(w), V^* \rangle(x)$.
- Hybrid $H_2^{V^*}(\lambda, x, w)$: This is identical to $H_1^{V^*}(\lambda, x, w)$ except that, for [Step 1](#) of the almost-total puzzle, it selects a random $i^* \in [n]$, and instead of sample x_{i^*}' randomly, it sends $x_{i^*}' = x_0^* \oplus x$ to the verifier. Since x_0^* is chosen at random, the two hybrid executions remain indistinguishable from the verifier's perspective. Thus, we have

$$\{H_1^{V^*}(\lambda, x, w)\}_{\lambda, x, w} \stackrel{s}{\approx} \{H_2^{V^*}(\lambda, x, w)\}_{\lambda, x, w} \quad (21)$$

where $\lambda \in \mathbb{N}$, $x \in L \cap \{0, 1\}^\lambda$, $w \in R_L(\lambda)$.

- Hybrid $H_3^{V^*}(\lambda, x, w)$: This is identical to $H_2^{V^*}(\lambda, w)$ except that, for the last round, $\mathcal{S}$ uses its non-uniform advice x_0^* (solution for the almost-total puzzle) as the witness in WIPoK. By the (non-uniform) WI property of WIPoK, we have

$$\{H_2^{V^*}(\lambda, x, w)\}_{\lambda, x, w} \stackrel{c}{\approx} \{H_3^{V^*}(\lambda, x, w)\}_{\lambda, x, w} \quad (22)$$

where $\lambda \in \mathbb{N}$, $x \in L \cap \{0, 1\}^\lambda$, $w \in R_L(\lambda)$. We remark that we can rely on the WI property against non-uniform adversaries by treating the interaction prior to the last round, as well as the verifier's internal state at the beginning of the last round, as non-uniform advice.

- Hybrid $H_4^{V^*}(\lambda, x, w)$: This is identical to $H_3^{V^*}(\lambda, w)$ except that it commits to 0 instead of the witness. By the (non-uniform) hiding property of Com, we have

$$\{H_3^{V^*}(\lambda, x, w)\}_{\lambda, x, w} \stackrel{c}{\approx} \{H_4^{V^*}(\lambda, x, w)\}_{\lambda, x, w} \quad (23)$$

where $\lambda \in \mathbb{N}$, $x \in L \cap \{0, 1\}^\lambda$, $w \in R_L(\lambda)$.

One can see that $H_4^{V^*}(\lambda, x, w)$ corresponds to the view simulated by $\mathcal{S}$. Thus, the computational indistinguishability between $H_1^{V^*}(\lambda, x, w)$ and $H_4^{V^*}(\lambda, x, w)$ directly implies the delayed-input non-uniform zero-knowledge property. This completes the proof of [Lem. 8](#). $\square$

7 Post-Quantum Public-Coin Extractable Commitment with EQPT_c-Simulation

We present in [Prot. 4](#) a *post-quantum public-coin extractable commitment* from *one-way functions* in five rounds. This protocol makes use of the following building blocks:

- A public-coin post-quantum preprocessing zero-knowledge argument of knowledge ppZKAoK with EQPT_c-simulation constructed in [Prot. 2](#). Its preprocessing stage is two-round and prove stage is three-round.
- Naor's commitment Com; We use β to denote the first message for Naor's commitment, and $\text{Com}_\beta(m; r)$ to denote the second message where a string m is committed using randomness r .

Protocol 4: Five-Rond Public-Coin PQ-ExtCom with EQPT _c -Simulation
Let m be the message to be committed to. Commit Stage: 1. (Preprocessing Stage of ppZKAoK.) Committer C and receiver R run the preprocessing stage of ppZKAoK where C and R play the roles of the prover and verifier, respectively. Let pp be the transcript of the preprocessing stage. 2. (Commit to Message) C commits to a message m using Naor's commitment. Formally, R samples and sends the first message β for Naor's commitment; C samples a random tape r and sends $c = \text{Com}_\beta(m; r)$. 3. (Prove Stage of ppZKAoK.) C and R execute the prove stage of ppZKAoK where C proves to R that he “knows” the committed message m in c. Formally, C proves that $c \in \mathcal{L}_\beta$ where $\mathcal{L}_\beta := \{c \mid \exists(m, r) \text{ s.t. } c = \text{Com}_\beta(m; r)\}. \quad (24)$ Decommit Stage: C sends (m, r). R accepts if $c = \text{Com}_\beta(m; r)$, and rejects otherwise.

Theorem 7. *Prot. 4 is a five-round public-coin post-quantum extractable commitment with EQPT_c-simulation (as per Def. 18).*

Proof. Correctness and the public-coin property are clear from the description of the protocol in Prot. 4. One can see that it runs in five rounds noting that the first and second messages in Step 2 can be sent parallel to the final message of Step 1 and the first message of Step 3, respectively. The statistical binding property immediately follows from that of Naor’s commitment.

To show the computational hiding property, we recall that the preprocessing zero-knowledge implies preprocessing strong witness indistinguishability (Lem. 7). The computational hiding is reduced to preprocessing strong witness indistinguishability in a straightforward manner since commitments to different messages are computationally indistinguishable by the computational hiding of Naor’s commitment and the commitment is sent to R after finishing the preprocessing stage of ppZKAoK.

Extractability with EQPT_c-simulation immediately follows from the adaptive AoK with EQPT_c-simulation property of ppZKAoK. $\square$

Theorem 8. *Prot. 4 is a post-quantum extractable commitment with rewinding-safe binding property (as per Def. 21).*

Proof (Sketch). The binding message is the second message in Step 2, and extractability for Prot. 4 relies on the adaptive AoK with EQPT_c-simulation property of ppZKAoK in Step 3. Therefore, during extraction, binding message is not rewound. $\square$

8 Post-Quantum Public-Coin Non-Malleable Commitment with Constant Tag

We now construct a *public-coin* PQ-NMCom with constant number of tags from *one-way function*. For the sake of simplicity, in the rest of the paper (Sec. 8 and 9), we are going to ignore the difference between fully-extractable and ε -extractable commitments in our proof of security. This is due to the fact that any (arbitrarily small) noticeable simulation error $\varepsilon(\lambda)$ does not affect the proof of non-malleability if we take the noticeable error to be much smaller than the assumed MIM adversary’s advantage.

8.1 Construction of PQ-NMCom with Constant Tag

This protocol makes use of the following building blocks:

- An almost-total puzzle constructed in Prot. 1.
- A post-quantum constant-round public-coin witness-indistinguishable argument WIAoK *without* post-extraction simulation (defined in Def. 10) constructed in Prot. 2.¹⁷
- Constant-round public-coin ε -extractable commitment scheme ExtCom with rewinding-safe binding property (defined in Def. 21) constructed in Prot. 4.¹⁸

¹⁷ ppZKAoK with EQPT_c-simulation implies WIAoK without post-extraction simulation

¹⁸ Our construction actually achieves negligible-error extraction under EQPT_c model, and by Lem. 1, it implies ε -extraction

Block Commitment. A block commitment $\mathcal{B}$ is a sequence of extractable commitments, and define

$$\text{val}(\mathcal{B}) := \begin{cases} v_1 \oplus \dots \oplus v_n & \text{if all extractable commitments are valid, and } v_i := \text{val}(\text{ExtCom}_i) \\ \perp & \text{otherwise} \end{cases}$$

where n is the length of the sequence, namely the number of extractable commitments in the sequence.

Protocol 5: Post-Quantum Public-Coin Non-Malleable Commitment with Constant Tag
Security parameter $\lambda \in \mathbb{N}$. r is the robustness parameter, k is the total number of messages in the extractable commitment protocol. T is the total number of tags. l is the maximum between (1) the robustness parameter r, and (2) the total number of messages in WIAoK. Commit Stage: (under an identification tag $\text{tg} \in [T]$) Phase 1: Commitments to Secret Shares of Value: - Committer C and receiver R run the almost-total puzzle protocol, with C playing the role of the puzzle receiver R_{PUZ} and R acting as the puzzle sender G_{PUZ}. If the protocol completes without any party aborting, they jointly generate a valid transcript τ; - Let $n := (k+1)^{\text{tg}}$ and $\bar{n} := (k+1)^{2T-\text{tg}}$. Committer C secret shares message $m \in \{0,1\}^*$ twice, first into n shares (using n out of n secret sharing), and second into $\bar{n}$ shares (using $\bar{n}$ out of $\bar{n}$ secret sharing). $u = (u_1, \dots, u_n)$ and $\bar{u} = (\bar{u}_1, \dots, \bar{u}_{\bar{n}})$, respectively. C provides extractable commitments to the two sequences of shares: (Block commitment $\mathcal{B}_1$ with n shares.) An extractable commitment to u_i, for every $i \in [n]$, sequentially, one after the other. (Block commitments $\mathcal{B}_{2 \dots l+2}$ with $\bar{n}$ shares.) An extractable commitment to $\bar{u}_i$, for every $i \in [\bar{n}]$, sequentially, one after the other. The sequential commitment to $\bar{u}$ is repeated $l+1$ times, sequentially. Define $\mathcal{B} := \{\mathcal{B}_1, \mathcal{B}_2, \dots, \mathcal{B}_{l+2}\}$ Phase 2: Proof of Consistency: C and R execute an instance of WIAoK^a where C proves to R that either he “knows” a solution for τ (defined in Step 1) or all block commitments commits to the same message m (defined in Step 3). Formally, C proves that $(\tau, \mathcal{B}) \in \mathcal{L}_\tau \vee \mathcal{L}_\mathcal{B}$, and $\mathcal{L}_\tau \vee \mathcal{L}_\mathcal{B}$ denotes the OR-composed language (Def. 1), where $\mathcal{L}_\tau := \{\tau \mid \exists s \text{ s.t. } \text{Verify}(\tau, s) = 1\}. \quad (25)$ and $\mathcal{L}_\mathcal{B} := \{\mathcal{B} \mid \exists m \text{ s.t. } \forall i \in \{1, \dots, l+2\}, \text{val}(\mathcal{B}_i) = m\}. \quad (26)$ Note that honest C commits to the same message m in all $l+2$ block commitments, and uses all the shares together with the corresponding decommitments as the witness for WIAoK. R accepts the commitment iff the proof of consistency is accepting. Decommit Stage: C sends all shares of the first block commitment $(u_1 \dots u_n)$ and the decommitment for each corresponding extractable commitment. R accepts if all decommitments are valid, and rejects otherwise.

^a Note that this WIAoK does not need post-extraction simulation (Def. 10)

8.2 MIM Adversaries with Predetermined Ideal Schedule

8.2.1 Adaptive Schedule vs. Predetermined Schedule

Similar to [BLS22, LPY23], we restrict attention to MIM adversaries that always announce ahead of time its schedule, and show our protocol is secure against all MIM adversaries with predetermined schedule. The following lemma established that if our protocol is secure against all predetermined schedule adversaries, then it is secure against arbitrary adversaries.

Lemma 13 ([LPY23, Lemma 11]). *Let $\mathcal{S}$ be the set of all possible schedules of a MIM adversary against a commitment scheme $\langle C, R \rangle$. Let $S_1, \dots, S_N$ be efficiently recognizable subsets of $\mathcal{S}$ such that $\bigcup_{i \in [N]} S_i = \mathcal{S}$ for $N = \text{poly}(\lambda)$. If for every $i \in [N]$, $\langle C, R \rangle$ is non-malleable against non-uniform QPT MIM adversaries whose schedule belongs to S_i , then it is also non-malleable against arbitrary non-uniform QPT MIM adversaries.*

8.2.2 Ideally Scheduled Block Commitments.

We first recall definition of ideally scheduled block commitments from [BLS22] with slight modifications to adapt our protocol. We then argue that, after our modifications, such ideally scheduled block commitment still exists. We first define *free* (block) commitment, in a two-sided MIM setting.

Definition 27 (Free (block) commitment). *We say a commitment B on the left (on the right) is free with respect to protocol A on the right (on the left), if it satisfies the following:*

- **Freeness** *From the start till the end of the execution of B , no message of protocol A is sent.*

Remark 7. In Def. 27, at a high-level, we require a free (block) commitment B to not interleave with A and can not be “split” by execution of A . But a free commitment B can “split” execution of A , and by freeness requirement, it can “split” at most one execution of A .

In a two-sided MIM execution of Prot. 5, we say an execution of a block commitment $\mathcal{B}_{j \in [\ell+2]}$ on the left is *free* on index i with respect to a $\mathcal{B}_{i^*}$ on the right, to mean that the i -th ExtCom of $\mathcal{B}_j$ is free with respect to a $\mathcal{B}_{i^*}$.

Definition 28 (Ideally Scheduled Block Commitment). *A block commitment $\mathcal{B}_{i^*}$ (for some $i^* \in [\ell+2]$) on the right is ideally scheduled, if it satisfies the following:*

1. *It is free with respect to the WIAoK on the left.*
2. *There is some index $i \in [n]$ such that the first left block commitment $\mathcal{B}_1$ is free on index i with respect to $\mathcal{B}_{i^*}$.*
3. *There is some index $j \in [\bar{n}]$ such that all the left block commitments $\mathcal{B}_2, \dots, \mathcal{B}_{\ell+2}$ on the left are free on the same index j with respect to $\mathcal{B}_{i^*}$.*

In case the MIM adversary aborts, we assume w.l.o.g it keeps sending messages $\perp$ according to some schedule, so that the above notion is always defined.

Remark 8. Compared with [BLS22], we replaced their ZK argument with WIAoK, and use our almost-total puzzle generator (Prot. 1) to setup a trapdoor at the beginning of the protocol. And in our modified definition of ideally scheduled block, we allow the right ideally scheduled block $\mathcal{B}_{i^*}$ to interleave with the left almost-total puzzle at the beginning of Prot. 5 but not the left WIAoK. Therefore the exact proof for existence of an ideally scheduled block from [BLS22] directly applies here.

Lemma 14 ([BLS22, Claim 5.1.]). *In every MIM execution of Prot. 5 with tag $\mathbf{tg}$ on the left and tag $\tilde{\mathbf{tg}}$ on the right, if $\mathbf{tg} \neq \tilde{\mathbf{tg}}$, there is an ideally scheduled execution of a block commitment on the right.*

8.2.3 Predetermined Ideal Schedule

Now we discuss all the possible predetermined ideal schedules for Prot. 5, we adopt the notation from [BLS22].

Let N be a bound on the size of $n := (k+1)^{\mathbf{tg}}$, $\bar{n} := (k+1)^{(2T-\mathbf{tg})}$, for every possible $\mathbf{tg}$. We consider *configurations* of the form

$$C = (i^*, c, \bar{c}, w) \in [l+2] \times [N] \times [N] \times \{\text{IP}_2, \text{P}_2\text{I}\}.$$

We say that a given MIM execution is consistent with such a configuration C if:

1. For any $i^* \in [l+2]$, the i^* -th block commitment $\mathcal{B}_{i^*}$ on the right is the first ideally scheduled block.
2. For any $c \in [N]$, the commitment to u_c (in the first block) on the left is free with respect to the ideally scheduled block $\mathcal{B}_{i^*}$.
3. For any $\bar{c} \in [N]$, the commitment to $\bar{u}_{\bar{c}}$ in every one of the blocks $2, \dots, l+2$ on the left is free with respect to the ideally scheduled block $\mathcal{B}_{i^*}$.
4. If $\mathcal{B}_{i^*}$ ends before Phase 2 on the left begins, $w = \text{IP}_2$. Otherwise $w = \text{P}_2\text{I}$.

Remark 9. For $w = \text{P}_2\text{I}$, by definition, it means that Phase 2 on the left begins before $\mathcal{B}_{i^*}$ ends. By the definition of ideally scheduled block, $\mathcal{B}_{i^*}$ on the right is continuous with respect to Phase 2 on the left (during its execution, no WIAoK message is sent on the left), we can also say that $\mathcal{B}_{i^*}$ on the right begins after Phase 2 on the left has started (rather than say that it only ends after the beginning of Phase 2 on the left).

8.3 Proof of Non-Malleability

Security. Statistical-binding and public-coin properties are straightforward from the description of Prot. 5. Computational-hiding property of any non-malleable commitment scheme follows directly from its non-malleability. Therefore we are left to prove the non-malleability and r -robustness of our protocol. Non-malleability is established by the following theorem.

Theorem 9. *The protocol described in Prot. 5 is non-malleable against any non-uniform QPT MIM adversary $\mathcal{A}^* = \{\mathcal{A}_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$.*

Proof. Following Lem. 13, we restrict attention to a MIM adversary with a predetermined ideal schedule $C = (i^*, c, \bar{c}, w)$. Specifically, since either $w = \text{P}_2\text{I}$ or $w = \text{IP}_2$, we show that the MIM experiment output distributions for any m_0, m_1 on the left are indistinguishable for both cases.

8.3.1 $w = \text{P}_2\text{I}$

If $w = \text{P}_2\text{I}$. Namely, Phase 2 on the left begins before the *ideally scheduled* block commitment $\mathcal{B}_{i^*}$ on the right had started. We consider the following sequence of hybrids.

Hybrid $H_0^{\text{P}_2\text{I}}(\lambda, m_b)$: This is the real experiment where C commits to m_b on the left, where $b \in \{0, 1\}$. It outputs the final output of MIM adversary $\mathcal{A}^*$ and $\tilde{m}$ committed on the right.

Hybrid $H_1^{P_2I}(\lambda, m_b)$: This is identical to $H_0^{P_2I}(\lambda, m_b)$ except that it outputs message $\tilde{m}_{i^*}$ committed in $\mathcal{B}_{i^*}$ (ideally scheduled) instead of $\tilde{m}$.

Claim 10. $H_1^{P_2I}(\lambda, m_b) \stackrel{s}{\approx} H_0^{P_2I}(\lambda, m_b)$

Proof. By the knowledge soundness of WIAoK, for any accepting WIAoK on the right, either MIM $\mathcal{A}^*$ knows a solution for the almost-total puzzle or MIM $\mathcal{A}^*$ commits to the same message $\tilde{m}$ in every block commitment on the right. Since our almost-total puzzle is hard (Property 1) against any QPT puzzle receiver, MIM $\mathcal{A}^*$ will not be able to find a solution for the almost-total puzzle, and thus commits on the right consistently with overwhelming probability. More formally, using a standard reduction, we build $\mathcal{A}'$ for the almost-total puzzle. Consider an external challenger for the almost-total puzzle; $\mathcal{A}'$ forward messages between the challenger and $\mathcal{A}$ on the right, and simulates an honest C on the left. If $\mathcal{A}^*$ does not abort, τ is a valid transcript on the right; If $\mathcal{A}'$ extract from the right WIAoK produced by $\mathcal{A}$, it finds a solution for τ and leads to a contradiction. Thus, given an accepting WIAoK, all block commitments on the right contains the same message $\tilde{m}$, and thus $\tilde{m}_{i^*} = \tilde{m}$ with overwhelming probability. $\square$

Hybrid $H_2^{P_2I}(\lambda, m_b)$: This is identical to $H_1^{P_2I}(\lambda, m_b)$ except that it invokes the (sequentially) ε -simulatable extractor $\mathcal{SE}_\varepsilon$ to obtain the all corresponding shares in $\mathcal{B}_{i^*}$. By the definition of ε -simulatable extractability (Def. 19), we have

$$H_2^{P_2I}(\lambda, m_b) \stackrel{c}{\approx} H_1^{P_2I}(\lambda, m_b)^{19} \quad (27)$$

Hybrid $H_3^{P_2I}(\lambda, m_b)$: This is identical to $H_2^{P_2I}(\lambda, m_b)$ except that it finds the lexicographically smallest solution s for the *left* almost-total puzzle by *brute-force*. If a solution does not exist, it outputs $\perp$, and otherwise works similarly to the previous hybrid. We note that this experiment obtains s but does not use it. By the almost-total property (Property 2), the probability that this hybrid outputs $\perp$ is negligible. Thus, we have

$$H_3^{P_2I}(\lambda, m_b) \stackrel{s}{\approx} H_2^{P_2I}(\lambda, m_b) \quad (28)$$

Hybrid $H_4^{P_2I}(\lambda, m_b)$: This is identical to $H_3^{P_2I}(\lambda, m_b)$ except that it uses the solution s (defined in Step 1) as the witness for the left WIAoK in phase 2.

Claim 11. $H_4^{P_2I}(\lambda, m_b) \stackrel{c}{\approx} H_3^{P_2I}(\lambda, m_b)$

Proof. In case $w = P_2I$, the ideally scheduled block $\mathcal{B}_{i^*}$ on the right starts after phase 2 on the right had started, and by the definition of ideally scheduled block, no WIAoK message can “split” $\mathcal{B}_{i^*}$. Therefore, invocation of the sequentially-extracting simulator $\mathcal{SE}$ does not compromise WI property of WIAoK. Therefore, by the non-uniform WI property of WIAoK, we have

$$H_4^{P_2I}(\lambda, m_b) \stackrel{c}{\approx} H_3^{P_2I}(\lambda, m_b)$$

$\square$

¹⁹ As we mentioned earlier, it is fine to ignore the ε simulation error in the proof of non-malleability

Hybrid $H_{5,j}^{P_2I}(\lambda, m_b)$ for $j = l+3, \dots, 1$: $H_{5,l+3}^{P_2I}(\lambda, m_b)$ is defined the same as $H_4^{P_2I}(\lambda, m_b)$. In what follows, recall that MIM $\mathcal{A}^*$ follows predetermined ideal schedule $C = (i^*, c, \bar{c}, w)$. For all $j \in [l+2]$, $H_{5,j}^{P_2I}(\lambda, m_b)$ is identical to $H_{5,j+1}^{P_2I}(\lambda, m_b)$ except that it changes the *free* extractable commitment of $\bar{u}_{\bar{c}}$ (u_c if $j = 1$) in the j -th block commitment on the left to an extractable commitment of 0.

Claim 12. $H_{5,j+1}^{P_2I}(\lambda, m_b) \stackrel{c}{\approx} H_{5,j}^{P_2I}(\lambda, m_b)$

Proof. Similar to the proof of [Thm. 11](#), $\mathcal{B}_{i^*}$ on the right starts after phase 2 on the right had started, which means that it starts after all $l+2$ block commitments have finished. So no commitment in $\mathcal{B}_{i^*}$ interleaves with or can be “split” by any left commitments. Therefore, invocation of the sequentially-extracting simulator does not compromise hiding property of ExtCom. By the non-uniform hiding property of ExtCom, we have,

$$H_{5,j+1}^{P_2I}(\lambda, m_b) \stackrel{c}{\approx} H_{5,j}^{P_2I}(\lambda, m_b)$$

□

Note that in $H_{5,1}^{P_2I}(\lambda, m_b)$ is independent of b , by the same sequence of hybrids (in reverse order), we have $H_0^{P_2I}(\lambda, m_0) \stackrel{c}{\approx} H_0^{P_2I}(\lambda, m_1)$ conditioned on $w = P_2I$.

8.3.2 $w = IP_2$

If $w = IP_2$. Namely, Phase 2 on the left begins after $\mathcal{B}_{i^*}$ (ideally scheduled) on the right ends. We consider the following sequence of hybrids.

Hybrid $H_0^{IP_2}(\lambda, m_b)$: This is the real experiment where C commits to m_b on the left, where $b \in \{0, 1\}$. It outputs the final output of MIM adversary $\mathcal{A}^*$ and $\tilde{m}$ committed on the right.

Hybrid $H_1^{IP_2}(\lambda, m_b)$: This is identical to $H_0^{IP_2}(\lambda, m_b)$ except that it outputs message $\tilde{m}_{i^*}$ committed in $\mathcal{B}_{i^*}$ instead of $\tilde{m}$. By the same argument as in the proof of [Thm. 10](#), we have

$$H_1^{IP_2}(\lambda, m_b) \stackrel{s}{\approx} H_0^{IP_2}(\lambda, m_b) \tag{29}$$

Hybrid $H_2^{IP_2}(\lambda, m_b)$: This is identical to $H_1^{IP_2}(\lambda, m_b)$ except that it finds the lexicographically smallest solution s for the *left* almost-total puzzle by *brute-force*. If a solution does not exist, it outputs $\perp$, and otherwise works similarly to the previous hybrid. We note that this experiment obtains s but does not use it. By the almost-total property ([Property 2](#)), the probability that this hybrid outputs $\perp$ is negligible. Thus, we have

$$H_2^{IP_2}(\lambda, m_b) \stackrel{s}{\approx} H_1^{IP_2}(\lambda, m_b) \tag{30}$$

Hybrid $H_3^{IP_2}(\lambda, m_b)$: This is identical to $H_2^{IP_2}(\lambda, m_b)$ except that it uses the solution s (defined in [Step 1](#)) as the witness for the left WIAoK in phase 2.

Claim 13. $H_3^{IP_2}(\lambda, m_b) \stackrel{c}{\approx} H_2^{IP_2}(\lambda, m_b)$

Proof. At a high level, we want to rely on WI property of the WIAoK (the left-session) to prove the indistinguishability between $H_3^{IP_2}(\lambda, m_b)$ and $H_2^{IP_2}(\lambda, m_b)$. However, recall that the output of the hybrid is defined to be the tuple $(\text{st}_{\mathcal{A}^*}, \tilde{m}_{i^*})$. Importantly, $\tilde{m}_{i^*}$ is *not* efficiently computable. This forbids an *efficient* reduction to the WI property. Fortunately, notice that $H_2^{IP_2}(\lambda, m_b)$ and $H_3^{IP_2}(\lambda, m_b)$ are identical until the beginning of the left-session WIAoK ([Step 4](#)), and thus the

distribution of $\tilde{m}_{i^*}$ is actually identical in $H_2^{\text{IP}_2}(\lambda, m_b)$ and $H_3^{\text{IP}_2}(\lambda, m_b)$. Therefore, the only source of detectable difference between $H_2^{\text{IP}_2}(\lambda, m_b)$ and $H_3^{\text{IP}_2}(\lambda, m_b)$ would be the final state $\text{st}_{\mathcal{A}^*}$ of the MIM adversary, which is *efficiently* computable. Then, the indistinguishability between the two hybrids follows from a standard reduction to the WI property where the distinguisher only gets $\text{st}_{\mathcal{A}^*}$ as the input. $\square$

Hybrid $H_{4,j}^{\text{IP}_2}(\lambda)$ for $j = l+3, \dots, 1$: $H_{4,l+3}^{\text{IP}_2}(\lambda)$ is defined the same as $H_3^{\text{IP}_2}(\lambda, m_b)$. In what follows, recall that MIM $\mathcal{A}^*$ follows predetermined ideal schedule $C = (i^*, c, \bar{c}, w)$. For all $j \in [l+2]$, $H_{3,j}^{\text{IP}_2}(\lambda)$ is identical to $H_{3,j+1}^{\text{IP}_2}(\lambda)$ except that it changes the *free* extractable commitment of $\bar{u}_{\bar{c}}$ (u_c if $j = 1$) in the j -th block commitment on the left to an extractable commitment of 0.

Claim 14. $H_{4,j}^{\text{IP}_2}(\lambda) \stackrel{c}{\approx} H_{4,j+1}^{\text{IP}_2}(\lambda)$

Proof. Let's call the *free* ExtCom in the j -th block commitment on the left $\text{ExtCom}_j^{\text{left}}$. At a high-level, we want to rely on the non-uniform hiding property of $\text{ExtCom}_j^{\text{left}}$. But again, $\tilde{m}_{i^*}$ committed in $\mathcal{B}_{i^*}$ on the right is *not* efficiently computable. This forbids an *efficient* reduction to the hiding property.

In order to solve this, we first define $H'_{4,j+1}(\lambda)$, which is identical to $H_{4,j+1}^{\text{IP}_2}(\lambda)$ except it extracts $\tilde{m}_{i^*}$ differently. In $H'_{4,j+1}(\lambda)$, all right ExtComs in $\mathcal{B}_{i^*}$ that end strictly before $\text{ExtCom}_j^{\text{left}}$ are fixed *non-uniformly*. For all right ExtComs in $\mathcal{B}_{i^*}$ that begin strictly after $\text{ExtCom}_j^{\text{left}}$, we invoke the (sequentially) ε -simulatable extractor $\mathcal{SE}_\varepsilon$. Note that $\text{ExtCom}_j^{\text{left}}$ can split at most one ExtCom in $\mathcal{B}_{i^*}$ on the right (by the definition of free Def. 27). Here we rely on the *rewinding-safe binding* property (Def. 21) of our ExtCom, which says that the first sender message that binds the value of the commitment occurs before extraction gadget of ExtCom. And we call such message the *binding* message. If binding message occurs before the commitment $\text{ExtCom}_j^{\text{left}}$ has started, we can non-uniformly fix it. And if binding message occurs after the commitment $\text{ExtCom}_j^{\text{left}}$ has ended, we invoke sequentially-extracting simulator. By the definition of ε -simulatable extractability (Def. 19), we have

$$H'_{4,j+1}(\lambda) \stackrel{c}{\approx} H_{4,j+1}^{\text{IP}_2}(\lambda)^{20} \quad (31)$$

We then define $H'_{4,j}(\lambda)$, which is identical to $H'_{4,j+1}(\lambda)$ except it changes $\text{ExtCom}_j^{\text{left}}$ to a commitment of 0. By the non-uniform hiding property of ExtCom, we have

$$H'_{4,j}(\lambda) \stackrel{c}{\approx} H'_{4,j+1}(\lambda) \quad (32)$$

By the same argument as above, we have

$$H_{4,j}^{\text{IP}_2}(\lambda) \stackrel{c}{\approx} H'_{4,j}(\lambda) \quad (33)$$

Therefore, Combining Eq. (31) to (33), we obtain Thm. 14. $\square$

Note that in $H_{4,1}^{\text{IP}_2}(\lambda, m_b)$ is independent of b , by the same sequence of hybrids (in reverse order), we have $H_0^{\text{IP}_2}(\lambda, m_0) \stackrel{c}{\approx} H_0^{\text{IP}_2}(\lambda, m_1)$ conditioned on $w = \text{IP}_2$. Combining with previous result, we conclude that Prot. 5 is non-malleable against all QPT MIM adversaries with predetermined schedule, and therefore non-malleable against all QPT adversaries. $\square$

²⁰ Again, we are ignoring ε extraction error

8.4 r -Robustness

Recall that in [Prot. 5](#), the second sequence of ExtComs are repeats $l + 1$ times (with $l \geq r$). This allows a safe extraction from any of the $l + 1$ sequence in [Prot. 5](#) without rewinding a r -message protocol on the right. We observe that the modifications we made in [Prot. 5](#) (compared to the small tag commitment scheme in [\[BLS22\]](#)) does not affect any of the rewinding opportunity, and therefore we can use their result directly.

Lemma 15 ([\[BLS22, Proposition 5.2.\]](#)). *The non-malleable commitment scheme from [Prot. 5](#) is extractable with r -robustness.*

9 Public-Coin Tag Amplification

In this section, we present a modified construction of [\[BLS22\]](#) tag amplification compiler, which will preserve *public-coin* and r -robustness property of the base non-malleable commitment. In more details, this compiler converts any non-malleable commitment scheme $\langle C, R \rangle$ for $t \in [3, O(\log(\lambda))]$ bit tags into a non-malleable commitment scheme $\langle \hat{C}, \hat{R} \rangle$ for $T = 2^{t-1}$ bit tags.

9.1 Construction of Public-Coin Tag Amplification Compiler

This compiler makes use of the following building blocks:

- An almost-total puzzle generator constructed in [Prot. 1](#).
- Naor’s commitment Com; We use β to denote the first message for Naor’s commitment, and $\text{Com}_\beta(m; r)$ to denote the second message where a string m is committed using randomness r .
- A post-quantum constant-round public-coin witness-indistinguishable argument WIAoK *without* post-extraction simulation (defined in [Def. 10](#)) constructed in [Prot. 2](#).²¹ Let r be the number of rounds in this WIAoK.
- A public-coin non-malleable commitment scheme $\langle C, R \rangle$ for $t \in [3, O(\log(\lambda))]$ bit tags that is r -robust.

The transformed non-malleable commitment scheme $\langle \hat{C}, \hat{R} \rangle$ has $T = 2^{t-1}$ bits tag.

Protocol 6: Public-Coin Tag Amplification Compiler
Commit Stage: (under an identification tag $\widehat{\text{tg}} \in \{0, 1\}^T$) 1. Trapdoor Setup: Committer C and receiver R run the almost-total puzzle protocol, with C playing the role of the puzzle receiver R_{PUZ} and R acting as the puzzle sender G_{PUZ}. If the protocol completes without any party aborting, they jointly generate a valid transcript τ; 2. Initial Commitment: Committer C commits to a message m using Naor’s commitment. Formally, Receiver R samples and sends the first message β for Naor’s commitment; C samples a random tape r and sends $\text{c} = \text{Com}_\beta(m; r)$; 3. Parallel Execution of $\langle C, R \rangle$: For every bit $\widehat{\text{tg}}_i$ in the $T = 2^{t-1}$ bit tag $\widehat{\text{tg}}$, define tag $\text{tg}_i := (i, \widehat{\text{tg}}_i)$, which has exactly t bits.

²¹ ppZKAoK with EQPT_c-simulation implies WIAoK without post-extraction simulation

For every $i \in [T]$, C commits to m using $\langle C, R \rangle$ under tag tg_i . let $\mathbf{c}_i$ denote the produced commitment and r_i the decommitment. All commitments are sent in parallel.

4. **Proof of Consistency:** C and R execute an instance of WIAoK^a where C proves to R that either he “knows” a solution for τ (defined in [Step 1](#)) or $\mathbf{c}, \mathbf{c}_1, \dots, \mathbf{c}_T$ all commits to the same message m . Formally, let $\vec{\mathbf{c}} := (\mathbf{c}, \mathbf{c}_1, \dots, \mathbf{c}_T)$, and $\vec{r} := (r, r_1, \dots, r_T)$. C proves that $(\tau, \vec{\mathbf{c}}) \in \mathcal{L}_\tau \vee \mathcal{L}_\beta$, and $\mathcal{L}_\tau \vee \mathcal{L}_\beta$ denotes the OR-composed language, where

$$\mathcal{L}_\tau := \{\tau \mid \exists s \text{ s.t. } \text{Verify}(\tau, s) = 1\}. \quad (34)$$

and

$$\mathcal{L}_\beta := \{\vec{\mathbf{c}} \mid \exists m, \vec{r} \text{ s.t. } \mathbf{c} = \text{Com}_\beta(m; r) \wedge \forall i \in \{1, \dots, T\}, \mathbf{c}_i = \text{Com}_\beta(m; r_i)\}. \quad (35)$$

Note that honest C commits to the same message m in $\mathbf{c}, \mathbf{c}_1, \dots, \mathbf{c}_T$, and uses $(m, \vec{r})$ as the witness for WIAoK . R accepts the commitment iff the proof of consistency is accepting.

Decommit Stage: C sends decommitment r . Decommitment is accepted if $\mathbf{c} = \text{Com}_\beta(m; r)$, and rejects otherwise.

^a Note that this WIAoK doesn't need post-extraction simulation ([Def. 10](#))

9.2 Proof of r -Robustness

In order to show r -robustness of $\langle \hat{C}, \hat{R} \rangle$, we will reduce it to the r -robustness of $\langle C, R \rangle$.

Theorem 15. *The protocol $\langle \hat{C}, \hat{R} \rangle$ described in [Prot. 6](#) is r -robust.*

Proof. Consider the following sequence of hybrids.

Hybrid H_0 : This is the real experiment and outputs $\text{OUT}_{\hat{\mathcal{A}}}(\langle B(1^\lambda, z), \hat{\mathcal{A}}^{\mathcal{O}[\langle \hat{C}, \hat{R} \rangle]}(\hat{\rho}) \rangle)$

Hybrid H_1 : This is identical to H_0 except that the oracle $\mathcal{O}[\langle \hat{C}, \hat{R} \rangle]$ is replaced by a wrapper oracle $\hat{\mathcal{O}}[\langle C, R \rangle]$. The wrapper oracle $\hat{\mathcal{O}}[\langle C, R \rangle]$ work as follow: it emulates the honest receiver $\hat{R}$ for $\hat{\mathcal{A}}$ except for the first commitment $\mathbf{c}_1$ (under tag tg_i) of $\langle C, R \rangle$ in [Step 3](#). For $\mathbf{c}_1$, $\hat{\mathcal{O}}[\langle C, R \rangle]$ forwards it to $\mathcal{O}[\langle C, R \rangle]$, which is the committed value oracle for $\langle C, R \rangle$. If a non-aborting $\hat{\mathcal{A}}$ requests **break**, our wrapper oracle $\hat{\mathcal{O}}$ will request **break** to $\mathcal{O}[\langle C, R \rangle]$ and obtain m_1 committed in $\mathbf{c}_1$ and send it to $\hat{\mathcal{A}}$.

Claim 16. $H_1 \stackrel{s}{\approx} H_0$

Proof. In H_0 , the honest receiver $\hat{R}$ for $\hat{\mathcal{A}}$ is emulated by $\mathcal{O}[\langle \hat{C}, \hat{R} \rangle]$. In H_1 , the honest receiver $\hat{R}$ for $\hat{\mathcal{A}}$ is emulated by our wrapper oracle $\hat{\mathcal{O}}[\langle C, R \rangle]$ (together with its call to $\mathcal{O}[\langle C, R \rangle]$). So both hybrids emulate the honest receiver $\hat{R}$ perfectly. Next, we show that on request **break**, value returned by $\hat{\mathcal{O}}[\langle C, R \rangle]$ and $\mathcal{O}[\langle C, R \rangle]$ are the same with overwhelming probability. Recall that $\hat{\mathcal{O}}[\langle C, R \rangle]$ returns value $\tilde{m}'$ committed in Naor's commitment $\mathbf{c}$ and $\mathcal{O}[\langle C, R \rangle]$ returns value $\tilde{m}_1$ committed in $\mathbf{c}_1$. Similar to the proof of [Thm. 10](#), by the knowledge soundness of WIAoK , for any accepting WIAoK on the right, either MIM $\mathcal{A}^*$ knows a solution for the almost-total puzzle or MIM $\mathcal{A}^*$ commits to the same message $\tilde{m}$ in every commitment on the right. Since our almost-total puzzle is hard ([Property 1](#)) against any QPT puzzle receiver, MIM $\mathcal{A}^*$ won't be able to find a solution for the almost-total puzzle, and thus $\tilde{m}_1 = \tilde{m}'$ with overwhelming probability. $\square$

Hybrid H_2 : This is identical to H_1 except that we view the output $\text{OUT}_{\hat{\mathcal{A}}}(B(z, 1^\lambda), \hat{\mathcal{A}}^{\mathcal{O}[\langle C, R \rangle]}(\hat{\rho}))$ differently. Instead of view it as an output of $\hat{\mathcal{A}}$ with oracle access to the wrapper oracle $\hat{\mathcal{O}}$, in hybrid H_2 , we view the output as $\text{OUT}_{\mathcal{A}}(B(z, 1^\lambda), \mathcal{A}^{\mathcal{O}[\langle C, R \rangle]}(\hat{\rho}))$, which is the output of a wrapper adversary $\mathcal{A}$ with oracle access to $\mathcal{O}[\langle C, R \rangle]$. The wrapper adversary $\mathcal{A}$ runs $\hat{\mathcal{A}}(\hat{\rho})$ internally. On the left session, it forwards messages between B and $\hat{\mathcal{A}}(\hat{\rho})$. On the right session, $\mathcal{A}$ behaves exactly the same as the wrapper oracle $\hat{\mathcal{O}}$ we defined in hybrid H_1 , it emulates the honest receiver $\hat{R}$ for $\hat{\mathcal{A}}(\hat{\rho})$ and forwards the first commitment c_1 to $\mathcal{O}[\langle C, R \rangle]$. If a non-aborting $\hat{\mathcal{A}}$ requests **break**, $\mathcal{A}$ will request **break** to $\mathcal{O}[\langle C, R \rangle]$ and obtain m_1 committed in c_1 and send it to $\hat{\mathcal{A}}(\hat{\rho})$. By definition, we have $H_2 = H_1$.

Hybrid H_3 : This is identical to H_2 except that instead of invoking the committed value oracle $\mathcal{O}[\langle C, R \rangle]$, it uses the simulator Sim_r to simulate output of $\mathcal{A}$. By the r -robustness of the underlying protocol $\langle C, R \rangle$, we have $H_3 \stackrel{c}{\approx} H_2$. Note that in H_3 , we construct a simulator $\widehat{\text{Sim}}_r$ for $\langle \hat{C}, \hat{R} \rangle$ such that.

$$\{\text{OUT}_{\widehat{\text{Sim}}_r}(B(1^\lambda, z), \widehat{\text{Sim}}_r(1^\lambda, \hat{\mathcal{A}}, \hat{\rho}))\}_{\lambda \in \mathbb{N}, z \in \{0,1\}^*} \stackrel{c}{\approx} \{\text{OUT}_{\hat{\mathcal{A}}}(B(1^\lambda, z), \hat{\mathcal{A}}^{\mathcal{O}[\langle \hat{C}, \hat{R} \rangle]}(\hat{\rho}))\}_{\lambda \in \mathbb{N}, z \in \{0,1\}^*}$$

Thus, protocol $\langle \hat{C}, \hat{R} \rangle$ is r -robust. □

9.3 Proof of Non-Malleability

Security. Statistical-binding and public-coin properties are straightforward from the description of [Prot. 6](#). Computational-hiding property of any non-malleable commitment scheme follows directly from its non-malleability. Therefore we are left to prove the non-malleability which is established by the following theorem.

Theorem 17. *The protocol described in [Prot. 6](#) is non-malleable against any non-uniform QPT MIM adversary $\mathcal{A}^* = \{\mathcal{A}_\lambda^*, \rho_\lambda\}_{\lambda \in \mathbb{N}}$.*

Proof. Similar to the base protocol [Prot. 5](#), we restrict attention to a MIM adversary with a predetermined schedule. We consider the following two cases:

- Case 1: If [Step 3](#) of the *right* session starts before [Step 1](#) of the *left* session ends.
- Case 2: If [Step 3](#) of the *right* session starts *after* [Step 1](#) of the *left* session ends.

According to [Lem. 13](#), we only need to show that the MIM experiment output distributions for any m_0, m_1 on the left are indistinguishable for both cases.

9.3.1 Case 1

In case 1, [Step 3](#) of the right commitment starts before [Step 1](#) of the *left* commitment ends. We consider the following sequence of hybrids.

Hybrid H_0^b : This is the real experiment where C commits to m_b on the left, where $b \in \{0, 1\}$. It outputs the final output of MIM adversary $\mathcal{A}^*$ and $\tilde{m}$ committed on the right.

Hybrid H_1^b : This is identical to H_0^b except that it uses *brute-force* to get message $\tilde{m}'$ committed in [Step 2](#) on the right. If it gets no message, it outputs $\perp$, and otherwise works similarly to the previous hybrid. We note that this experiment obtains $\tilde{m}'$ but does not use it.

Claim 18. $H_1^b \stackrel{s}{\approx} H_0^b$

Proof. Similar to the proof of [Thm. 10](#), by the knowledge soundness of WIAoK, for any accepting WIAoK on the right, either MIM $\mathcal{A}^*$ knows a solution for the almost-total puzzle or MIM $\mathcal{A}^*$ commits to the same message $\tilde{m}$ in every commitment on the right. Particularly, if $\mathcal{A}^*$ commits to the same message $\tilde{m}$ in every commitment on the right, then the Naor's commitment ([Step 2](#)) sent by MIM $\mathcal{A}^*$ must be a valid commitment, namely a commitment that binds a value with valid decommitment. Since our almost-total puzzle is hard ([Property 1](#)) against any QPT puzzle receiver, MIM $\mathcal{A}^*$ will not be able to find a solution for the almost-total puzzle, and thus commits on the right consistently with overwhelming probability. Therefore, $\tilde{m}'$ exists with overwhelming probability, and the probability that this hybrid outputs $\perp$ is negligible. $\square$

Hybrid H_2^b : This is identical to H_1^b except that it outputs message $\tilde{m}'$ instead of $\tilde{m}$. Recall that $\tilde{m}$ is the committed message on the right, which is defined to be the message committed in Naor's commitment in [Prot. 6](#). And $\tilde{m}'$ is the message we brute-forced from Naor's commitment. Thus, we have

$$H_2^b = H_1^b \quad (36)$$

Hybrid H_3^b : This is identical to H_2^b except that it finds the lexicographically smallest solution s for the almost-total puzzle ([Step 1](#)) by *brute-force*. If a solution does not exist, it outputs $\perp$, and otherwise works similarly to the previous hybrid. We note that this experiment obtains s but does not use it. By the almost-total property ([Property 2](#)), the probability that this hybrid outputs $\perp$ is negligible. Thus, we have

$$H_3^b \stackrel{s}{\approx} H_2^b \quad (37)$$

Hybrid H_4^b : This is identical to H_3^b except that it uses the solution s ([Step 1](#)) as the witness for the WIAoK for proof of consistency ([Step 4](#)).

Claim 19. $H_4^b \stackrel{c}{\approx} H_3^b$

Proof. Observe that H_4^b and H_3^b proceed identically before [Step 4](#) on the left starts, and $\tilde{m}'$ is fixed before [Step 2](#) on the left starts. So, in the output of H_4^b and H_3^b , $\tilde{m}'$ is identical. Let $\text{ST}_{\mathcal{A}^*}^{H_4^b}$ be the internal state of $\mathcal{A}^*$ of H_4^b , and $\text{ST}_{\mathcal{A}^*}^{H_3^b}$ be the internal state of $\mathcal{A}^*$ of H_3^b .

At a high-level, we want to rely on the non-uniform WI of the left WIAoK, and fortunately, both hybrids' output are efficiently computable. Recall that in H_4^b and H_3^b , $\tilde{m}'$ is identical and taken out by brute-force, therefore can be used as a non-uniform advice. This allows an *efficient* reduction to WI. By the non-uniform WI property of WIAoK, we have

$$(\text{ST}_{\mathcal{A}^*}^{H_4^b}, \tilde{m}') \stackrel{c}{\approx} (\text{ST}_{\mathcal{A}^*}^{H_3^b}, \tilde{m}') \quad (38)$$

By definition of non-malleability, this is equivalent to

$$H_4^b \stackrel{c}{\approx} H_3^b \quad (39)$$

More formally, using a standard reduction, we build $\mathcal{A}'$ for the WI property of WIAoK. Consider an external challenger for WI; On the right, $\mathcal{A}'$ simulates an honest R with MIM $\mathcal{A}^*$. On the left, $\mathcal{A}'$ simulates an honest C except for [Step 4](#) on the left. In more details, $\mathcal{A}'$ use brute-force to get $\tilde{m}'$

and solution s for [Step 1](#) on the left. and then sends s together with m_0 and all the decommitment $(r, r_1, \dots, r_T)$ to the challenger (for [Steps 2](#) and [3](#) on the left). For the left WIAoK, $\mathcal{A}'$ forward messages between the challenger and $\mathcal{A}^*$. In the end, $\mathcal{A}'$ sends the final state of $\mathcal{A}^*$ ($\text{ST}_{\mathcal{A}^*}$) and $\tilde{m}'$ to the distinguisher $\mathcal{D}$. Observe that, depending on the choice of the outside challenger, $\mathcal{A}'$ emulates one of the hybrids perfectly. Thus, for any QPT distinguisher $\mathcal{D}$, if $\mathcal{D}$ distinguishes H_4^b from H_3^b , then $\mathcal{A}'$ breaks the hiding property of the left WIAoK and leads to a contradiction. Thus, H_4^b and H_3^b are computationally indistinguishable. $\square$

Hybrid H_5^b : This is identical to H_4^b except that it commits to 0 in the left Naor's commitment ([Step 2](#)). Similar to the proof of [Thm. 19](#), instead this time we rely on the non-uniform hiding property of Naor's commitment, we have

$$H_5^b \stackrel{c}{\approx} H_4^b \quad (40)$$

Hybrid $H_{6:i}^b$ for $i \in \{0, 1, \dots, T\}$: Hybrid $H_{6:0}^b$ is identical to H_5^b . For $i \in [T]$, hybrid $H_{6:i}^b$ is identical to $H_{6:i-1}^b$ except that in [Step 3](#), it commits to 0 in the i -th base non-malleable commitment $\langle C, R \rangle$. Similar to the proof of [Thm. 19](#), instead this time we rely on the non-uniform hiding property of $\langle C, R \rangle$, we have

$$H_{6:i}^b \stackrel{c}{\approx} H_{6:i-1}^b \quad (41)$$

Note that in $H_{6:T}^b$ is independent of b , by the same sequence of hybrids (in reverse order), we have $H_0^0 \stackrel{c}{\approx} H_0^1$.

9.3.2 Case 2

In case 2, [Step 3](#) of the right commitment starts *after* [Step 1](#) of the *left* commitment ends. We consider the following sequence of hybrids.

Hybrid H_0^b : This is the real experiment where C commits to m_b on the left, where $b \in \{0, 1\}$. It outputs the final output of MIM adversary $\mathcal{A}^*$ and $\tilde{m}$ committed on the right.

Hybrid H_1^b : This is identical to H_0^b except that it outputs message $\widetilde{m_{i^*}}$ committed in i_{th}^* commitment $\widetilde{c_{i^*}}$ in [Step 3](#) on the right, for an i^* that $\text{tg}_{i^*} \neq \widetilde{\text{tg}_{i^*}}$. Similar to the proof of [Thm. 18](#), we have

$$H_1^b \stackrel{s}{\approx} H_0^b \quad (42)$$

Hybrid H_2^b : This is identical to H_1^b except that it finds the lexicographically smallest solution s for the almost-total puzzle ([Step 1](#)) by *brute-force*. If a solution does not exist, it outputs $\perp$, and otherwise works similarly to the previous hybrid. We note that this experiment obtains s but does not use it. By the almost-total property ([Property 2](#)), the probability that this hybrid outputs $\perp$ is negligible. Thus, we have

$$H_2^b \stackrel{s}{\approx} H_1^b \quad (43)$$

Hybrid H_3^b : This is identical to H_2^b except that it uses the solution s ([Step 1](#)) as the witness for the left WIAoK for proof of consistency ([Step 4](#)).

Claim 20. $H_3^b \stackrel{c}{\approx} H_2^b$

Proof. We consider the following sequence of hybrids. $\mathcal{G}_1^b$: This is identical to H_2^b except that it invokes the (sequentially) simulatable extractor $\mathcal{SE}_r$ to get $\widetilde{m_{i^*}}$. Since our base protocol $\langle C, R \rangle$ is r -robust extractable, it is also simulatable-extractable. By the definition of simulatable extractability (Def. 19), we have

$$\mathcal{G}_1^b \stackrel{c}{\approx} H_2^b \quad (44)$$

$\mathcal{G}_2^b$: This is identical to $\mathcal{G}_1^b$ except that it uses the solution s (Step 1) as the witness for the left WIAoK for proof of consistency (Step 4). By the non-uniform WI property of WIAoK, and the r -robustness property of $\langle C, R \rangle$ (Def. 20), we have

$$\mathcal{G}_2^b \stackrel{c}{\approx} \mathcal{G}_1^b \quad (45)$$

H_3^b : This is identical to $\mathcal{G}_2^b$ except that it doesn't get $\widetilde{m_{i^*}}$ by rewinding, and outputs $\widetilde{m_{i^*}}$ directly (inefficient). Again, by the simulation extraction property, we have

$$H_3^b \stackrel{c}{\approx} \mathcal{G}_2^b \quad (46)$$

□

Hybrid H_4^b : This is identical to H_3^b except that it commits to 0 in the left Naor's commitment (Step 2). By a similar argument for Thm. 20, from the non-uniform hiding property of Naor's commitment and the r -robustness property of the base protocol, we have

$$H_4^b \stackrel{c}{\approx} H_3^b \quad (47)$$

Hybrid $H_{5:i}^b$ for $i \in \{0, 1, \dots, T\}$: Hybrid $H_{5:0}^b$ is identical to H_4^b . For $i \in [T]$, hybrid $H_{5:i}^b$ is identical to $H_{5:i-1}^b$ except that in Step 3, it commits to 0 in the i_{th} base non-malleable commitment. Recall that in the output of hybrid $H_{5:i}^b$ is set to be $\widetilde{m_{i^*}}$ together with $\text{ST}_{\mathcal{A}^*}^{H_5}$. Observe that for every $i \in [T]$, we have $\text{tg}_i \neq \widetilde{\text{tg}_{i^*}}$. By the non-malleability of $\langle C, R \rangle$, we have

$$H_{5:i}^b \stackrel{c}{\approx} H_{5:i-1}^b \quad (48)$$

Note that in $H_{5:T}^b$ is independent of b , by the same sequence of hybrids (in reverse order), we have $H_0^0 \stackrel{c}{\approx} H_0^1$, conditioned on case 2. Combining with previous result, we conclude that Prot. 6 is non-malleable against all QPT MIM adversaries with predetermined schedule, and therefore non-malleable against all QPT adversaries. □

By iteratively applying the above tag amplification transformation for $L = O(\log^* \lambda)$ times on the small-tag NMCom construction with constant-size tags, we obtain a scheme with exponential-size tags. Since our construction follows the same framework as [BLS22], we refer the reader to [BLS22] for a detailed analysis of the complexity growth and security loss.

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